

CHAPTER OUTLINE

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KEY POINTS

- Along with rapid economic development, diabetic nephropathy is fast becoming the leading cause of chronic kidney disease and end-stage kidney disease in many Asian countries. Data from major kidney biopsy registries show that immunoglobulin A nephropathy (IgAN) is the leading diagnosis among primary glomerular diseases and lupus nephritis is the leading diagnosis among secondary glomerulonephritides.
- Treatment of active severe lupus nephritis with corticosteroids and mycophenolate is effective and associated with fewer adverse effects compared with prior treatment regimens combining corticosteroids with cyclophosphamide; and recent data showed that maintenance therapy with low-dose corticosteroids and mycophenolate was associated with a relatively low flare rate.
- Results from recent major clinical trials in IgAN highlight the potential risks of high-dose corticosteroid treatment and the critical importance of judicious balance between potential benefit versus risk in treatment decisions.
- Experience with aristolochic acid nephropathy and endemic (Balkan) nephropathy highlights the importance of awareness of potential pathogenic contributions by environmental toxins in patients presenting with acute kidney injury or chronic kidney disease of known or unknown etiology.
- In contrast to Caucasian populations in which granulomatosis with polyangiitis and antiproteinase 3 antibodies are more common, antineutrophil cytoplasmic antibody-associated vasculitis in East and Northeast Asia is characterized by a high percentage of patients presenting with microscopic polyangiitis and antimyeloperoxidase antibodies and more chronic lesions in the kidney biopsy.
- In the current era with effective nucleoside/tide therapy for hepatitis B, the short- and medium-term survival of hepatitis B surface antigen (HBsAg)-positive kidney transplant recipient is comparable with those without hepatitis B virus infection.

INTRODUCTION

The term “Far East” refers to East Asia (including Northeast Asia), part of North Asia, and Southeast Asia (Fig. 80.1).^{1,2} In addition to its vast geographical extent and big population, Asia is characterized by marked variations in socioeconomic development, ethnicity, and culture, both between countries and within big countries. Such heterogeneity is associated

with marked differences in epidemiology of kidney diseases including both causes of acute kidney injury (AKI) and chronic kidney disease (CKD), and access to health care, the latter with direct implications on clinical outcomes especially patient and kidney survival rates. The disease pattern in developed areas such as Japan, Korea, Singapore, and Hong Kong largely resembles that of the West, with diabetic nephropathy being the major cause of end-stage kidney disease (ESKD) and immunoglobulin A (IgA) nephropathy being



Fig. 80.1 Countries and territories that constitute the Far East. (Modified from Wikipedia, https://en.wikipedia.org/wiki/Geography_of_Asia.)

the leading primary renal parenchymal disease. With rapid economic development, the major cities in China are moving toward the same pattern with diabetic nephropathy quickly becoming the leading cause of ESKD. Some conditions are more common in Asia, for example, IgA nephropathy (IgAN), lupus nephritis, chronic hepatitis B, and aristolochic acid (AA) nephropathy. Conditions that are typically found in the tropics or subtropics such as malaria or dengue fever could present as imported cases in other Asian countries, while AKI associated with infections such as leptospirosis,

scrub typhus, or hemorrhagic fever with renal syndrome (HFRS) due to hantavirus is still present in some countries. The massive volume of international travel and commuting by humans and traffic of commodities between nations, and the climate change associated with global warming have all contributed to the spread of certain diseases such as dengue fever to previously less afflicted areas.

While kidney diseases, especially AKI and CKD, constitute a major public health problem with increasing cost to communities globally, the data on disease varieties and burden,

and also on the kidney health-care structure and delivery systems for treatment and prevention, remain fragmented. The International Society of Nephrology (ISN) has undertaken the Global Kidney Health Atlas project, which was a multinational cross-section survey to gather information on the current capacity for kidney care over all world regions, and the findings showed marked diversity in nephrologist density and institutional kidney care provision within Asia and also considerable knowledge gaps in some Asian countries.³ For example, data on patients receiving maintenance dialysis are available only in countries or areas with national health insurance or renal registries such as Japan, Korea, Taiwan, Singapore, and Hong Kong, and few countries have data on AKI or nondialysis CKD. Renal replacement therapies are costly, and hemodialysis (HD) facilities may not be immediately available to patients with AKI in rural areas. The outcome of patients with AKI would thus be adversely affected due to financial disadvantage or inconvenient social setting. It is against this background that the ISN has launched the 0by25 initiative to prevent death related to AKI through the promulgation of knowledge and advocacy to facilitate early diagnosis and the building of sustainable capacities for timely clinical management.⁴ The benefits of this project extend beyond the areas of direct impact such as some South Asian countries to other parts of Asia, such as the less developed regions in China, through enhancing awareness and promoting proactivity and preparedness.

Health expenditure and health-care funding models, which have a direct impact on the access to health care, vary widely between Asian countries. Economic development varies markedly within the Far East region, ranging from high-income economies such as Japan, South Korea (Republic of Korea), Singapore, Hong Kong, and Taiwan, to low- or lower-middle income economies such as North Korea (Democratic People's Republic of Korea), the Philippines, Indonesia, Vietnam, Cambodia, Lao People's Democratic Republic, Myanmar, and Mongolia.⁵ Among Asian countries Japan has the highest health expenditure, amounting to 10.2% of gross domestic product (GDP) in 2014, compared with 17.1% in the United States of America.⁶ Health-care systems in places such as Japan, South Korea, Singapore, and Taiwan are based on national insurance and are well developed, while the health insurance coverage is being expanded in other areas such as China and Hong Kong.

TRADITIONAL MEDICINE

Many countries in Asia have a long history of traditional systems of medicine, and the use of traditional herbal medicine is still very common among Chinese in different countries and in other Asian countries such as Korea, Thailand, and Japan. Traditional medicines have served a significant role in providing health-care access to patients in less developed regions and in relatively low-income economies. For example, it has been reported that traditional Chinese medicine accounted for >20% of health-care services and treated approximately 200 million people annually in China.⁷ Traditional medicine is covered to variable extents by national health insurance schemes in China, Taiwan, South Korea, and Japan, and there are formal education systems for traditional medicine in China and Korea.⁸ Regulatory systems are in place in some areas, such as the State Traditional Chinese Medicine Administration

in China, the Ministry of Health and Welfare in Korea, the Ministry of Health and Welfare in Taiwan, and the Department of Health in Hong Kong, which include compliance requirements with regard to good manufacturing practice.⁹ However, due to the complexities in plant taxonomy and the components that are related to medicinal effects or toxicities, adverse effects related to intentional or inadvertent exposure to herbal medicines are not uncommon. Nephrotoxicity of *Tripterygium wilfordii*, a traditional Chinese medicine used in the treatment of lupus, rheumatoid diseases, and psoriasis, is well recognized,¹⁰ and the etiology, modes of exposure, and pathogenic mechanisms in AA nephropathy serve as illustrative examples.¹¹ A study in Taiwan reported a 20% increase in the risk of developing CKD in subjects who took herbal medicine regularly.¹² While the heritage of traditional medicine has its values and undeniable contributions, such as the case of artemisinin (from the plant qing hao) in the treatment of malaria, care and caution must be exercised especially in view of the commercial interest and its increasing use globally.

This chapter discusses the renal diseases that are more common, or of special interest, in the Far East, focusing on recent data and developments.

ACUTE KIDNEY INJURY AND CHRONIC KIDNEY DISEASE

ACUTE KIDNEY INJURY

AKI can be induced by a diverse range of endogenous or environmental factors. AKI affects >13 million people per year, although few countries maintain a record or have a registry for AKI patients.⁴ AKI is an important cause of hospital admission in Asian countries, and the leading causes of AKI vary according to the socioeconomic development status and the living standard of the general population in different countries.^{4,13} In areas with less developed building infrastructure and where the majority of the population is engaged in agriculture, community-acquired AKI is common and the causes include infections and animal or plant toxins (Table 80.1).¹⁴ Some of the causes are related to culture or folklore, for example, accidental exposure to toxin in the grass carp fish bladder used by merchants to replace the more costly snake gallbladder or ingestion of raw or dried cyprinid fish gallbladder as a health tonic or for the treatment of rheumatism,¹⁵ followed by the development of AKI which is sometimes accompanied by hepatotoxicity and myopathy due to the 5- α -cyprinol sulfate nephrotoxin/hepatotoxin.¹⁶ Another example of great public health importance is the exposure to AA present in herbs used in traditional medicine, which leads to not only CKD but also to AKI and multifocal urological malignancies.^{17,18} In more developed high-income countries such as Japan, Korea, or Singapore, causes such as toxin- or infection-associated AKI are much less common, while AKI is often a complicating factor in hospitalized critically ill patients with cardiovascular compromise, surgical conditions, and severe infections. In large countries with rapid economic development, such as China, the epidemiology of AKI varies according to geographical and topographical location.¹⁹ Knowledge of locally prevalent causes of AKI is thus crucial to timely therapeutic intervention.

Table 80.1 Secondary Renal Diseases Presenting as AKI in East, Southeast, and Northeast Asia

Causes	Examples
Autoimmunity	• Lupus nephritis • ANCA vasculitis • Antiglomerular basement membrane disease • Miscellaneous glomerulonephritis
Infections	• Leptospirosis • Dengue fever • Hantaviruses (hemorrhagic fever with renal syndrome) • Scrub typhus • Malaria • Severe fever with thrombocytopenia syndrome virus
Toxins	• From plant (e.g., aristolochic acid, djenkolic acid) • From animal or insect (e.g., snake venom, grass carp fish gallbladder) • Chemical (e.g., ketamine abuse presenting with CKD or AKI due to obstructive uropathy)

AKI, Acute kidney injury; ANCA, antineutrophil cytoplasmic antibody; CKD, chronic kidney disease.

CHRONIC KIDNEY DISEASE

CKD affects 10%–15% of the adult population worldwide.^{20–22} A meta-analysis reported in 2016 showed that the prevalence rate of Stage 1–5 CKD was approximately 13% in China, Japan, Korea, and Taiwan, whereas that for Stage 3–5 CKD was approximately 10%–12%.²¹ An earlier study in China reported a CKD prevalence rate of 10.8%, implying that approximately 120 million adults in China had reduced kidney function and/or albuminuria.²³ It was estimated that, in the year 2010 and for the whole of Asia, 218 patients per million population (pmp) were receiving dialysis and 232 patients pmp were receiving renal replacement therapy (RRT). In comparison, 1839 patients pmp were on renal replacement therapies in the United States in the same period.²⁴

A report from South Korea in 2016, based on the results of the Korean National Health and Nutritional Examination Survey in 2011–2013 that involved 23,280 subjects showed that the prevalence rate of CKD as defined by urine albumin-to-creatinine ratio ≥ 30 mg/g or estimated glomerular filtration rate (eGFR) < 60 mL/min/1.73 m² with the Chronic Kidney Disease-Epidemiology Collaboration equation was 8.2% in adults at or above the age of 20 years, with respective proportions in Stages 1, 2, 3a, 3b, and 4–5 being 3.0%, 2.7%, 1.9%, 0.4%, and 0.2%.²⁵ Factors associated with increased risk of CKD included older age, diabetes, hypertension, cardiovascular disease, body mass indexes ≥ 25 kg/m² or below 18.5 kg/m², and residence in a rural area. The KoreaN Cohort Study for Outcomes in Patients With Chronic Kidney Disease (KNOW-CKD) investigated the factors that impact the progression of CKD, based on data from 2238 subjects enrolled in 2011–2015.^{26,27} The age of the cohort was 53.7 ± 12.2 years and 61.2% were men. eGFR was 50.5 ± 30.3 mL/min/1.73 m²,

and lower eGFR values were associated with older age, comorbidities, higher systolic blood pressure, greater pulse pressure, lower income level, and lower education attainment. Glomerulonephritis, diabetic nephropathy, hypertensive nephropathy, and polycystic kidney disease accounted for 36.2%, 23.2%, 18.3%, and 16.3% of cases, respectively. The overall prevalence of previous cardiovascular disease was 14.4%, and the rate was highest in those with diabetes. The diabetic cohort also showed more adverse cardiovascular risk profiles including higher systolic blood pressure, higher levels of cardiac troponin T, higher left ventricular mass index, higher coronary calcium score, and higher brachial-ankle pulse wave velocity than the others.²⁷

The relationship between kidney function and survival was investigated in the population-based China National Survey of Chronic Kidney Disease study, which recruited adult subjects in 13 provinces starting from 2007.²³ Data from 47,204 subjects over a median follow-up duration of 6.1 years showed that all-cause and cardiovascular mortality increased with increasing severity of CKD and also albuminuria.²⁸ Baseline albumin-to-creatinine ratio ≥ 300 mg/g was associated with a hazard ratio (HR) of 2.07 for all-cause mortality and 2.32 for cardiovascular mortality compared with subjects with a ratio < 30 mg/g. Interestingly, reduced eGFR was not identified as a death risk predictor in multivariate analysis, although there was interaction between eGFR and age.²⁸

Health-care economics has a direct impact on access to clinical care and the availability and/or choice of therapies. This is particularly relevant to CKD care and renal replacement therapies, which are costly especially in Asia where the rate of organ transplantation is low and many patients remain on dialysis for years.^{3,29} Not surprisingly, large treatment gaps with regard to RRT have been noted in low-income countries in Asia.²⁴ Data from the International Society of Nephrology Global Kidney Health Care Status survey showed a variety of financing models ranging from total government funding to a mixture of public and private facilities for dialysis service in North and East Asia, with marked variations in the availability of services for CKD monitoring and management at the primary care level.^{3,30} In addition, very few countries have a registry for nondialysis CKD (or AKI) patients.^{3,31} In this context, a cost-effectiveness study from Japan published in 2015 showed that, based on 29 maintenance HD patients (eight had diabetes mellitus) with mean age of 59.9 years, the estimated total medical cost for HD treatment was approximately 45,200 USD annually, and the incremental cost utility ratio was $6.88 \pm 4.47 \times 10^4$ USD per quality-adjusted life-year, being higher for patients with diabetes mellitus ($8.17 \pm 6.28 \times 10^4$ USD) per quality-adjusted life-year.³² The challenge to health-care financing for CKD is compounded when economic development often brings about increasing life expectancy and a growing prevalence of CKD.²⁴ Heterogeneities in health-care spending, the developmental status of nephrology care and personnel, and sociocultural background result in marked variations in practice across Asian countries. For example, the health expenditure as a percentage of GDP in 2014 was 10.2% in Japan, highest among Asian countries, and was 7.373% in Korea, 5.548% in China, 4.710% in the Philippines, and 4.122% in Thailand, compared with 9.4% in Australia, 9.1% in the United Kingdom, 11.3% in Germany, and 17.1% in the United States.⁶ National health-care insurance in places such as Japan, Taiwan, Korea, and Singapore

provides coverage for dialysis therapy of residents. In places such as Hong Kong, Thailand, and Vietnam, economic or facility considerations have led to peritoneal dialysis (PD) being promoted or preferred to HD for the treatment of patients with ESKD as public health-care policies.^{33–38} Similarly, government-sponsored reimbursement schemes have led to an increase in PD utilization in some provinces in China, so that the number of PD patients in China increased from 37,942 in 2012 to 62,589 in 2015.^{39,40} Twice weekly or incremental HD has also been used in patients of small body size or with residual renal function to limit health-care expenditure.^{41,42}

In Japan, maintenance HD for patients with ESKD started in 1963.⁴³ A report from the Japanese Society for Dialysis Therapy Renal Data Registry in 2015, which covered 304,856 prevalent dialysis patients in 2011 (97% on HD), showed a progressive increase in the number of dialysis patients over the years, primarily due to increased number of patients above the age of 65 years, so that the mean age of prevalent dialysis patients was 66.6 years in 2011, compared with 46.2 years in 1981.^{44,45} Diabetes mellitus was the leading cause of ESKD, accounting for approximately 40% of incident patients and 45% of prevalent cases. By contrast, glomerular diseases accounted for 75% of incident patients in 1983, decreasing to around 35% in 2011. Cardiovascular disease was the leading cause for mortality, accounting for around 40%. In addition, approximately 30% of dialysis patients were above the age of 75 years, and 8% of patients had been on dialysis for >20 years.⁴⁴

While kidney transplantation offers the best potential for rehabilitation and is associated with lower long-term treatment costs, the current transplantation rate only meets approximately 10% of the demand globally.⁴⁶ Asia is the most populated region in the world but it also has the lowest rate of

organ transplantation.⁴⁶ Socioeconomic, cultural, political, legislative, and developmental issues contribute to the overall low organ donation rate, development of organ transplantation facilities, and access to such treatment.⁴⁷ Attempts to overcome some of the challenges have resulted in the development of kidney transplantation across ABO blood group barrier,⁴⁸ exchange donor kidney transplantation,⁴⁹ improvements in the management of hepatitis B after kidney transplantation so that hepatitis B virus (HBV)-infected renal failure patients can be included for kidney transplantation,^{50,51} and the use of hepatitis B surface antigen (HBsAg)-positive kidney donors.⁵²

SPECIFIC KIDNEY DISEASES

IGA NEPHROPATHY

IgAN is the most common primary glomerulonephritis worldwide, and the condition is common in Asia (Fig. 80.2).⁵³ Data from renal biopsy registries in China showed that IgAN accounted for >45% of primary glomerular diseases.^{54,55} The pathogenesis of IgAN is related to the overproduction of aberrantly glycosylated IgA1, which shows a relative deficiency of galactose at some hinge-region O-linked glycans, coupled with glycan-specific IgA and IgG autoantibodies that recognize the undergalactosylated IgA1 molecule.⁵³ Nephritogenic IgA1 molecules are produced by B cells following mucosal infections such as tonsillitis, and IgA and IgG that recognize the autoantigen IgA1 in IgAN are typically also reactive against antigens from microorganisms involved in recurrent upper respiratory and gastrointestinal mucosal infections. Glomerular deposition of immune complexes thus formed leads to activation of mesangial cells and overproduction of cytokines,

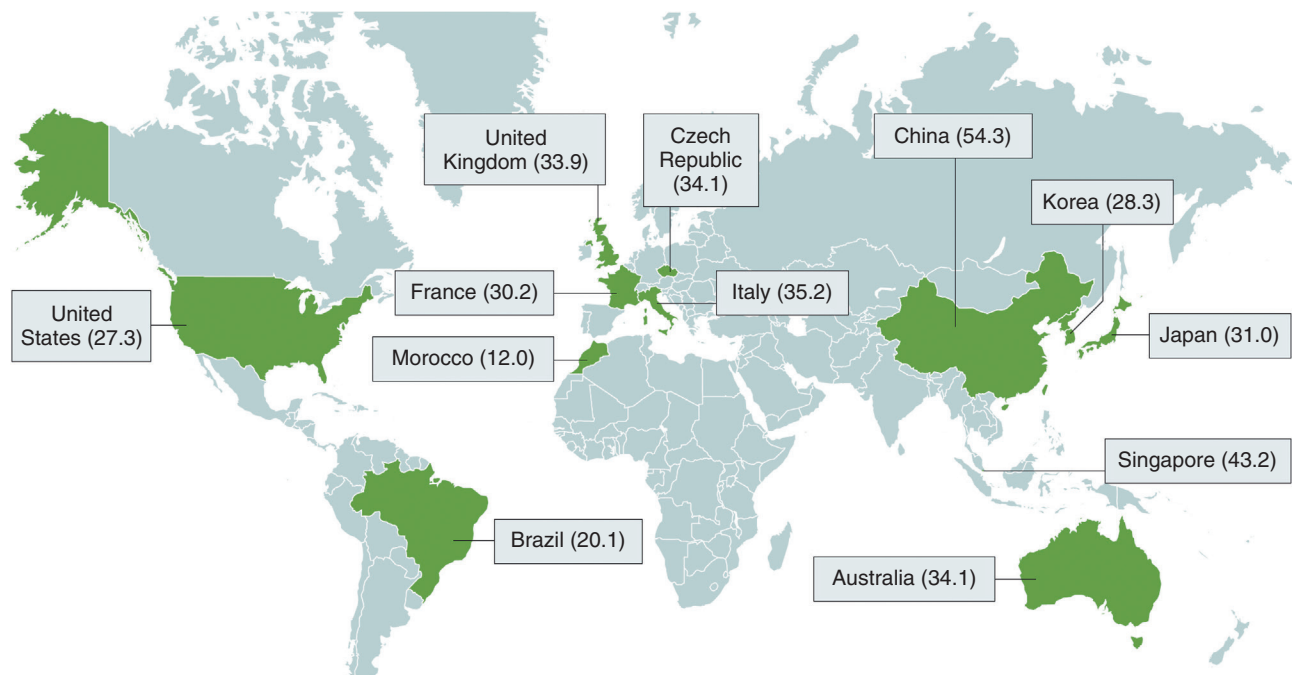


Fig. 80.2 Prevalence rates of immunoglobulin A nephropathy in different parts of the world. (Reprinted by permission from Macmillan Publishers Ltd. From Lai KN, Tang SCW, Schena FP, et al. IgA nephropathy. *Nat Rev Dis Primers* 2016;2:16001.)

chemokines, and other proinflammatory and profibrotic mediators locally, resulting in injury to normal parenchyma and replacement with fibrous tissue.

Results from linkage and genome-wide association study (GWAS) analyses have shown susceptibility loci of IgAN.^{56,57} Studies of familial IgAN in Chinese using whole-exome sequencing identified six deleterious variants in four genes, namely, defensin alpha 4 (*DEFA4*), MYC target 1 (*MYCT1*), caspase recruitment domain family member 8 (*CARD8*), and zinc-finger protein 543 (*ZNF543*).⁵⁸ Disease susceptibility has also been associated with genes involved in antigen processing and presentation such as the major histocompatibility complex locus or the mucosal defense system (the *DEFA* gene cluster), and also the alternative complement pathway.^{56,59}

Clinical presentation of IgAN is variable, ranging from asymptomatic microscopic hematuria with or without proteinuria or renal impairment to rapidly progressive renal impairment associated with abundant proliferative features and crescent formation in the glomeruli. Overall, about 30%–40% of patients reach ESKD in 20–30 years after initial presentation, but the prognosis and the rate of progression varies considerably between patients.⁵³ There is no specific blood or urine biomarker for the diagnosis of IgAN. The diagnosis is based on renal biopsy features and the detection of IgA in the mesangial area. Pathogenic polymeric IgA1 immune complexes, with IgG or IgM in some patients, deposit or form in situ in the glomerular mesangium and are associated with mesangial cell proliferation; increased synthesis of extracellular matrix; and variable infiltration of macrophages, monocytes, and T cells. The Oxford classification for IgAN was established in 2009 and updated in 2016 with the objective of identifying specific pathologic features that more accurately predict risk of progression of renal disease.^{60–62} According to the classification, renal biopsies of IgAN are scored according to mesangial hypercellularity (M0 or M1), segmental glomerulosclerosis (S0 or S1), endocapillary hypercellularity (E0 or E1), tubular atrophy/interstitial fibrosis (T0, T1, T2), and crescents (C0, C1, or C2) (MEST C Score), items that were shown to have independent value in predicting renal outcome. Analysis of pooled data from 3096 patients, including 700 patients from Japan and 1017 from China, showed that crescents predicted a higher risk of the combined renal endpoint, which was at least 50% eGFR decline or ESKD, in patients not treated with immunosuppression. Having crescents in at least one-sixth or one-fourth of glomeruli was associated with an HR of 1.63 or 2.29, respectively, for the combined renal endpoint, and having crescents in at least one-fourth of glomeruli was independently associated with a combined event irrespective of immunosuppression.⁶³ The T score in the Oxford classification and the Japanese Histologic Classification score have been shown to be associated with renal outcome in a series that included 86 Japanese patients with IgAN.⁶⁴

Based on the Oxford classification, the European Validation study of the Oxford classification of IgAN (VALIGA) study examined 1147 patients from 13 European countries encompassing different clinical and pathologic manifestations of IgAN.⁶⁵ Over a median follow-up of 4.7 years, 86% received renin–angiotensin–aldosterone system (RAAS) blockade and 42% immunosuppressive treatment. The results showed that M, S, and T lesions independently predicted the loss of eGFR and a lower renal survival. The addition of M, S, and T

lesions to clinical variables significantly enhanced the ability to predict progression only in patients not treated with immunosuppression.⁶⁵ In addition, applying the Oxford classification in 261 European patients below the age of 23 years with mean follow-up of 4.9 (range 2.5–8.1) years, investigators of the VALIGA cohort found that the rate of renal function decline was -0.2 mL/min/1.73 m² per year (interquartile range -2.7 to 0.5 mL/min/1.73 m² per year), and that the MEST M, S, and T scores predicted survival to the combined endpoint of ESKD and 50% loss of eGFR in the overall cohort but not in the subgroup of 174 children below the age of 18 years.⁶⁶ Applying the Oxford classification, investigators in Singapore showed that mesangial hypercellularity and tubular atrophy/interstitial fibrosis scores (M1 and T1/T2 lesion) of the Oxford classification independently predicted ESKD over long-term follow-up.⁶⁷

Treatment decisions take into account clinical and histological parameters. Patients with normal blood pressure, insignificant proteinuria, normal renal function, and renal histology demonstrating the absence of poor prognostic features may need only periodic monitoring of blood pressure, proteinuria, and renal function. Optimal treatment of hypertension and the use of RAAS inhibition or blockade, with the objective of minimizing proteinuria and the rate of renal function deterioration, form the mainstay of management strategy in other patients. The Kidney Disease Improving Global Outcomes (KDIGO) 2012 guidelines suggest optional use of fish oil in patients with persistent proteinuria of >1 g/day despite standard of care therapy, but its impact on long-term renal outcome remains uncertain.⁶⁸ A recent study in Japan showed that mizoribine treatment conferred no benefit on proteinuria.⁶⁹ The role of mycophenolate therapy remains controversial, and this is not recommended in the 2012 KDIGO guidelines, and the factors contributing to the conflicting conclusions from different studies include inadequate sample size, race and ethnicity, and histological variations.^{70–75}

Results from the Supportive Versus Immunosuppressive Therapy for the Treatment of Progressive IgA Nephropathy (STOP-IgAN) trial, which included 162 patients in Germany, showed that the addition of immunosuppressive therapy in the form of corticosteroids with or without cyclophosphamide or azathioprine in patients who showed persistent proteinuria of at least 0.75 g/day while on standard of care reduced proteinuria but did not reduce the rate of eGFR decline (24% in the supportive-care group and 22% in the immunosuppression group had a decrease in eGFR of at least 15 mL/min) >3 years of follow-up, and immunosuppression was associated with more infections (resulting in one death), impaired glucose tolerance, and weight gain.⁷⁶ The Therapeutic Evaluation of Steroids in IgA Nephropathy Global (TESTING) trial investigated the efficacy and safety of corticosteroids in 262 patients (251 patients in China) with IgAN and persistent proteinuria of >1 g/day despite standard therapy.⁷⁷ Corticosteroid treatment was in the form of oral methylprednisolone 0.6–0.8 mg/kg daily for 2 months then tapered by 8 mg/day each month, with a total treatment duration of 6–8 months. The trial was stopped early due to significantly higher incidence of serious adverse events in the methylprednisolone group (14.7% vs. 3.2%), primarily due to infections. In patients treated with methylprednisolone, 8.1% developed serious infections with two fatalities, whereas there was no serious infection in the placebo group. At trial

termination with median follow-up of 2.1 years, the primary efficacy outcome, which was a composite of ESKD, death due to kidney failure, or a 40% decrease in eGFR, favored methylprednisolone over placebo (5.9% vs. 15.9%; HR: 0.37).⁷⁷ In a trial that included 176 Chinese patients with IgAN and cellular/fibrocellular crescents in $\geq 10\%$ of glomeruli or endocapillary hypercellularity or glomerular necrosis, treatment with high-dose prednisone alone was comparable with mycophenolate combined with reduced-dose prednisone with regard to proteinuria reduction, and approximately 50% of patients in each group achieved complete renal remission with the latter group showing a lower incidence of treatment-related adverse effects.⁷⁸ Overall, the data from the trials that involved big samples of Chinese patients suggest potential renal benefit of immunosuppression, especially in patients whose renal histology shows active proliferative features, and that treatment decisions need to take into consideration the balance between risk and potential benefit according to the characteristics of individual patients. The data to date are insufficient to conclude that race or ethnicity has an impact on the response of IgAN to immunosuppressive treatments, due to the marked heterogeneity in histological manifestations and clinical disease severity, and the variable timing of diagnosis in the natural course of disease between different studies.

Microbial pathogens can multiply in the tonsillar crypts, and tonsillar lymphocytes of patients with IgAN showed increased production of dimeric IgA1 with altered glycosylation compared with controls.^{79,80} In addition, tonsillar germinal center B cells are implicated in pathogenesis.⁸¹ The role of tonsillectomy in the management of IgAN remains controversial. Tonsillectomy has been associated with more favorable renal outcome,^{82,83} and is considered a standard of care treatment for IgAN in Japan.^{84,85} A randomized controlled trial in Japan showed lower proteinuria in patients treated with tonsillectomy plus corticosteroids.⁸⁶ A recently reported long-term retrospective study in 227 Japanese patients showed possible benefit of tonsillectomy in patients presenting with lower levels of proteinuria and serum creatinine, but some of the patients also received corticosteroid treatment.⁸⁷ By contrast, analysis of data from the VALIGA cohort of 1147 European patients with IgAN showed no association between tonsillectomy and renal function decline over a follow-up of 4.7 years.⁸⁸

LUPUS NEPHRITIS

Lupus nephritis is common in Asia. Previous studies showed that not only is systemic lupus erythematosus (SLE) more common in Asian populations compared with Caucasians, but also renal involvement is more common among these patients.^{89,90} Data from renal biopsy registries showed that lupus nephritis is the leading cause of secondary glomerular diseases, accounting for $>50\%$ in countries such as China, Korea, and Japan.^{54,55,91–93}

A recent meta-analysis of GWAS data showed substantial commonality in shared risk variants between European and Chinese patients with SLE, with overall higher risk variant frequencies in the Chinese, suggesting a greater SLE genetic risk burden in East Asia.⁹⁴ Compared with other race or ethnicity, Asian patients may also develop more severe nephritis, resulting in lupus nephritis being an important cause of

ESKD in Asia.⁹⁵ Genetic association studies have identified >50 polymorphisms associated with lupus nephritis susceptibility or pathogenesis, encompassing functional abnormalities such as altered programmed cell death, defective immune clearance of programmed cell death debris, altered innate or adaptive immunity, and kidney-specific effector mechanisms.⁹⁶ An example is the reported association between variants of CCR6, a surface marker for Th17 effector cells, and susceptibility to lupus nephritis in a study that involved >1100 Chinese patients with lupus nephritis compared with 900 healthy controls and 1063 nonrenal lupus patients.⁹⁷

Clinical manifestation and the diagnosis of lupus nephritis in Asian patients are no different from other parts of the world.⁹⁸ However, within Asia there is marked variation in the timing of presentation due to variations in health-care delivery systems and primary care setup between countries. This has important implications on treatment and patient outcomes, because earlier presentation is in general associated with less severe disease that requires less immunosuppressive exposure for treatment. Earlier presentation is usually associated with less chronic irreversible damage. Reduced exposure to immunosuppressive medications, especially corticosteroids, leads to a more favorable efficacy versus risk balance and a lower risk of infective and other complications, whereas a lower level of damage in the kidneys implies a more favorable long-term renal survival. Except in some developing South Asian countries and disadvantaged populations in less developed areas in China and other Asian countries, the standard diagnostic facilities including blood and urine tests and lupus serology are available, though there may be economic barriers to kidney biopsy and regular serological tests. High baseline serum creatinine and significant chronic damage shown in the kidney biopsy portend an unfavorable long-term renal prognosis, whereas histologic features such as vascular or endothelial abnormalities and crescents signify aggressive disease.^{99–103} Studies from China have implicated antineutrophil cytoplasm antibodies in the pathogenesis of crescentic lupus nephritis,^{99,104,105} while a study in Korea reported that anti-Sm antibody was detected in 48.8% of patients with biopsy-proven lupus nephritis, and was associated with worse renal prognosis.¹⁰⁶ Furthermore, data from Korea showed that patients presenting after the age of 50 years showed higher renal chronicity features and had inferior outcomes during follow-up compared with those with disease onset at a younger age.¹⁰⁷ Renal histology of lupus nephritis is presently assessed according to the 2003 ISN/Renal Pathology Society classification,¹⁰⁸ although evaluation of individual lesions according to the specified diagnostic criteria may not be always easy.¹⁰⁹

While instituting prompt and effective treatment in Asian patients with severe lupus nephritis is intuitively important in view of the high disease prevalence in the region, this presents a challenge in some Asian regions due to insufficient knowledge of disease and management by patients, great variations in socioeconomic status of patients, variations in health-care financing and access to diagnosis and treatment, susceptibility to treatment-related adverse effects, access to and standard of general medical care and management of severe life-threatening complications, compliance, and follow-up issues.^{110–112}

Immunosuppressive treatment for lupus nephritis has evolved considerably since the 1980s, resulting in improved

clinical outcomes and significantly reduced treatment-associated adverse effects, and a shift from corticosteroids combined with cyclophosphamide to corticosteroids combined with mycophenolate therapy.¹¹³ Initial results demonstrating treatment efficacy and tolerability of combined prednisolone and mycophenolate therapy in Chinese lupus nephritis patients were subsequently confirmed by data from a Food and Drug Administration (FDA)-supported trial in the United States, data from a trial in Malaysia, and the industry-led international multicenter Aspreva Lupus Management Study, in which 123 of the 370 randomized patients were Asians.^{114–119} The reduced incidence of adverse events with mycophenolate treatment compared with cyclophosphamide was associated with improved quality of life.¹²⁰ Results from these trials showed racial variation in response to treatment, with inferior efficacy rates observed in patients of African descent and in Latin America while high response rates were observed in Chinese patients.^{114,115,121} The high short-term response rates in Chinese patients translated into relatively favorable long-term patient and renal survival, and are associated with a relatively low rate of nephritic flares.^{122–124}

Mycophenolic acid pharmacokinetics vary markedly between patients, and therapeutic drug monitoring is increasingly used in the management of organ transplant recipients treated with various preparations of mycophenolate.^{125–127} Studies in Thai patients with lupus nephritis showed that with the commonly adopted daily mycophenolate mofetil dose of 1.5–2 g the mycophenolic acid exposure of most patients was in the desired therapeutic range.^{128–130} However, while selected single time point drug levels correlated with area-under-curve data overall, drug exposure with the same single time point level can vary considerably between patients. In view of the generally high efficacy and tolerability in Asian lupus nephritis patients treated with mycophenolate, and the cost and availability of drug level monitoring, therapeutic drug level monitoring should preferably be reserved for patients who show unsatisfactory treatment response, including multiple relapses, or who develop marrow suppression, and in pediatric patients.¹³¹ The accumulating data on tacrolimus in the treatment of lupus nephritis have largely come from Asia. Owing to its distinct effect on podocytes in addition to potent immunosuppressive actions, calcineurin inhibitors have demonstrated efficacy in the treatment of lupus nephritis, especially when there were membranous features.^{132–139}

Investigators in China have reported the results with a ‘multitarget therapy’ with triple immunosuppression comprising corticosteroids, tacrolimus (4 mg/day), and reduced-dose (1.0 g/day) mycophenolate mofetil, showing that the multitarget regimen was associated with a higher short-term complete remission rate (65%) compared with corticosteroids and intravenous (IV) cyclophosphamide (15%) in patients with combined class IV and class V lupus nephritis,¹⁴⁰ a higher complete remission rate (45.9%) compared with IV cyclophosphamide (25.6%) at 24 weeks in patients with class III/IV/V lupus nephritis,¹⁴¹ and a similar renal flare rate (5.47%) compared with azathioprine (7.62%) as maintenance therapy for 18 months with the latter showing more adverse events.¹⁴²

Investigators in Japan have performed a pilot study in 15 patients with lupus nephritis to investigate another ‘multitarget’ immunosuppressive treatment regimen that included prednisolone, IV cyclophosphamide for 3 months, and tacrolimus at a standard dose of 3 mg daily, and reported a

complete remission rate of 80% at 6 months, with three patients showing a transient increase of serum creatinine level initially, one cytomegalovirus antigenemia, and one death related to thrombotic microangiopathy.¹⁴³ Overall, the results with tacrolimus in the treatment of lupus nephritis are promising, especially with regard to reduction of proteinuria. In view of potential nephrotoxicity which could be subclinical initially, and also its effects on blood pressure, lipids, and glucose metabolism, long-term safety data are required and further studies are warranted to delineate which are the patients most likely to benefit from this treatment.

In Japan, mizoribine is the first drug approved (in 1990) for the indication of lupus nephritis treatment. The mechanism of action is similar to mycophenolic acid, and the drug has been largely used during the maintenance therapy for steroid sparing, with generally acceptable tolerability. A recently reported postmarketing surveillance study on long-term mizoribine treatment in 559 patients with lupus nephritis showed that overall there was a continued decline in disease activity and 26.5% of patients were in complete remission at 24 months, yet progressive deterioration of kidney function was observed after 12 months of follow-up. Serious adverse drug reactions occurred in 3.6% of patients. In addition, nearly all patients were on concomitant corticosteroid treatment, whereas 51.2% of patients were also receiving tacrolimus.¹⁴⁴

The data on Asian patients with lupus nephritis show that the clinical outcomes with regard to response to treatment of active disease, long-term patient and renal survival, morbidities related to disease or treatment, disease flare rate, and quality of life of patients have improved over the past few decades due to advances in immunosuppressive management and improvements in general medical care. There are still questions and challenges with regard to optimal treatment regimens and durations for novel and existing therapies including calcineurin inhibitors, socioeconomic barriers to appropriate and timely diagnosis and treatment, and the prevention and management of infections.^{110,111}

DIABETIC NEPHROPATHY

Diabetic nephropathy, mostly due to type 2 diabetes in Asia, has become a major cause of ESKD in many Asian cities, in parallel with economic development. In Japan, diabetic nephropathy has overtaken glomerulonephritis as the leading underlying cause of ESKD in dialysis patients since 1997, and the proportion continues to increase (Fig. 80.3).⁴⁴ In China, among hospitalized patients, diabetic nephropathy has been the leading cause of CKD surpassing glomerulonephritis (Fig. 80.4).¹⁴⁵ The association between diabetes mellitus and CKD has similarly been demonstrated in Korea and Singapore.^{146,147} In a cross-sectional survey that included 1861 patients with type 2 diabetes in Singapore, 53% were found to have concomitant CKD, and variables significantly associated with CKD included retinopathy or neuropathy, age, blood pressure of $\geq 140/80$ mm Hg, history of cardiovascular disease, and duration of diabetes.¹⁴⁷

Around 30%–40% of type 2 diabetic patients develop diabetic nephropathy, the pathogenesis of which involves genetic and environmental factors.^{148,149} The genes related to the initiation and progression of diabetic nephropathy remain to be investigated. Previous studies have adopted the candidate gene approach, based on pathophysiologic

considerations, or GWASs. Data to date show that the RAAS plays an important role in the progression of diabetic nephropathy, and RAAS blockers are now standard therapy to reduce proteinuria and the rate of renal function deterioration.¹⁵⁰ Among candidate genes, genetic polymorphism of angiotensin-converting enzyme (ACE) gene has been studied intensively. The ACE gene 287-bp Alu repetitive sequences in intron 16 show genetic variations that create insertion

(II) or deletion (DD) homozygous or insertion/deletion (ID) heterozygous genotypes, and that the D allele is associated with high ACE enzymatic activity not only in sera but also in renal tissues, and data from single-center studies and meta-analyses showed that the D allele and DD genotype were associated with higher susceptibility to ESKD.^{151–153} In addition, an association with acetyl coenzyme A carboxylase β (ACACB) gene, which may be related to proinflammatory responses in renal tubular epithelial cells, has been reported in Japanese and Chinese patients.^{154,155} Considering the role of inflammation in pathogenesis, data from Chinese patients in Henan Province showed that the AA genotype and A allele of interleukin-10 (IL-10) –1082G/A was associated with increased susceptibility to diabetic nephropathy with adjusted odds ratios of 2.52 and 1.51 compared with wild-type genotype and the G allele, respectively.¹⁵⁶

Data from 568 Chinese patients with type 2 diabetes in Taiwan showed that IL-6 gene polymorphisms rs1800796 and rs1524107 predicted progression of nephropathy.¹⁵⁷ An association between diabetic nephropathy and intercellular adhesion molecule-1 (ICAM1) K469E(A/G) polymorphism (rs5498) has been reported in Malaysian patients.¹⁵⁸ In another study in Singapore that included 1950 Asians with type 2 diabetes, a higher plasma level of vascular cell adhesion molecule-1 (VCAM1), but not ICAM1, was associated with worse renal function and proteinuria.¹⁵⁹ With regard to the role of lipids in disease progression, higher levels of lipoprotein(a) has been associated with the development of CKD in Korean patients with type 2 diabetes,¹⁶⁰ and recent data from Chinese patients with type 2 diabetes suggested an association between diabetic nephropathy and apolipoprotein E isoforms, with the ApoE ϵ 2 allele being more prevalent and the ApoE ϵ 4 allele less prevalent in diabetic patients with nephropathy compared with those without renal abnormality.¹⁶¹ Dietary salt intake and serum uric acid level have been reported as independent risk factors, with odds ratios of 1.15 and 2.00,

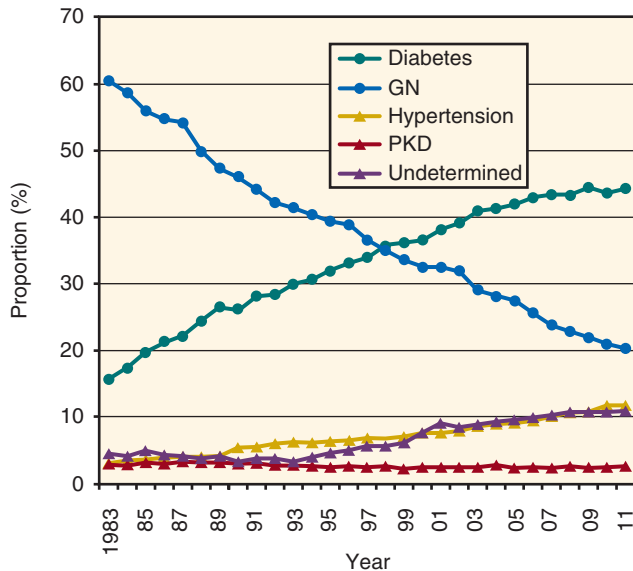


Fig. 80.3 Proportion of primary diagnoses of end-stage renal disease in prevalent patients on dialysis in Japan. GN, Glomerulonephritis; PKD, polycystic kidney disease. (Reprinted with permission from Elsevier. From Hanafusa N, Nakai S, Iseki K, et al. Japanese Society for Dialysis Therapy Renal Data Registry – a window through which we can view the details of Japanese dialysis population. *Kidney Int Suppl.* 2015;5:15–22.)

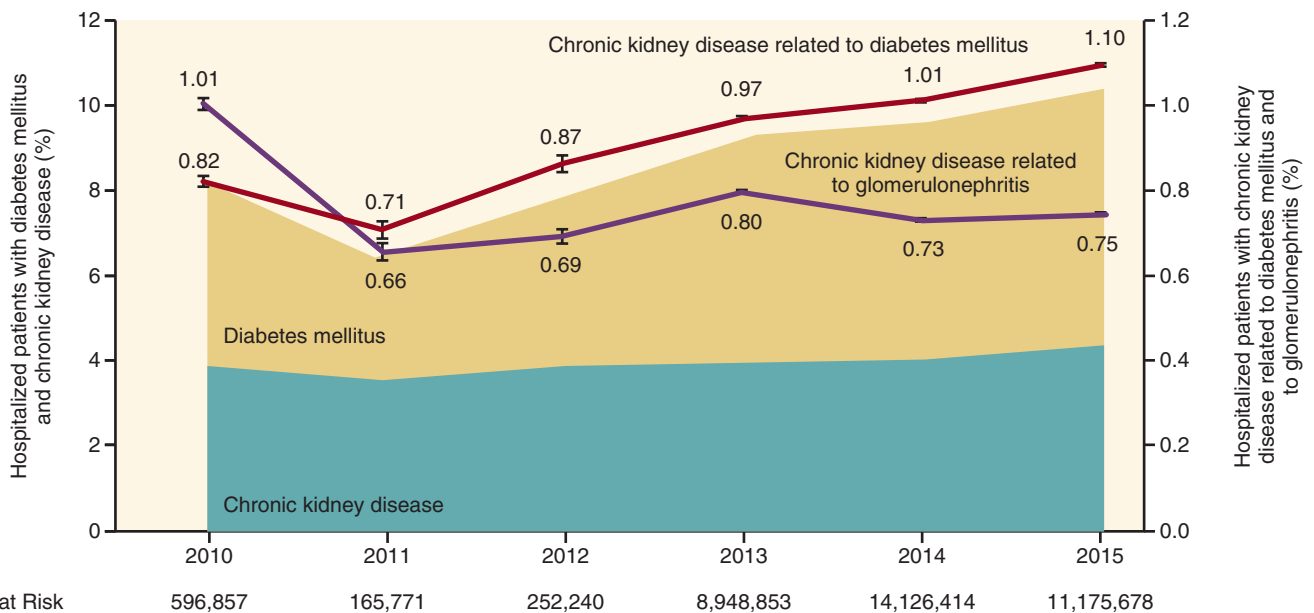


Fig. 80.4 Trends in chronic kidney disease related to diabetes and to glomerulonephritis among hospitalized patients in China. (Reprinted from Zhang L, Long J, Jiang W, et al. Trends in chronic kidney disease in China. *N Engl J Med.* 2016;375:905–906. Copyright (2016) Massachusetts Medical Society, with permission from the Massachusetts Medical Society.)

respectively, associated with renal impairment in Japanese patients with type 2 diabetes.¹⁶² Examples of other genetic polymorphisms implicated in susceptibility or progression of diabetic nephropathy include that of ELMO1 in Chinese,¹⁶³ 5,10-methylenetetrahydrofolate synthetase in Taiwan,¹⁶⁴ and 2184AG polymorphism in the gene for the receptor for advanced glycation end product in China.¹⁶⁵ In the field of epigenetics, data from Taiwan showed higher levels of miR-21, miR-29a, and miR-192 in diabetic patients with heavier proteinuria, and an association between miR-21, miR-29a, miR-29b, and miR-29c with progressive renal failure.¹⁶⁶

As in other parts of the world, Asian patients with diabetic nephropathy also show markedly increased cardiovascular risk and thus control of vascular risk factors is important in patient management.¹⁶⁷ Data from the West showed that combined use of ACE inhibitor (lisinopril) and angiotensin receptor blocker (losartan) compared with single-agent treatment may reduce proteinuria but did not reduce cardiovascular mortality and was potentially risky with regard to hyperkalemia and AKI.¹⁶⁸ Adding the direct renin inhibitor (aliskiren) to diabetic nephropathy patients already on angiotensin receptor blocker losartan reduced proteinuria and might better preserve the renal function.^{169,170} Subsequently, the Aliskiren Trial in Type 2 Diabetes Using Cardio-Renal Disease Endpoints (ALTITUDE) trial was stopped prematurely when interim analysis showed that adding aliskiren 300 mg daily to an ACE inhibitor or an angiotensin receptor blocker, though resulted in reduction of blood pressure and proteinuria, was associated with significantly higher occurrence rates of hyperkalemia of ≥ 6 mmol/L (11.2% vs. 7.2% in the placebo group) and hypotension (12.1% vs. 8.3% in controls; $P < .001$ for both).¹⁷¹ Caution is thus advised when considering dual blockade in the RAAS pathway. The balance between potential benefit versus risk is affected by the characteristics of individual patients.

In a study that included 563 patients with diabetic nephropathy from Japan and Hong Kong, with 73.5% already receiving treatment with an ACE inhibitor, treatment with the angiotensin receptor blocker olmesartan was shown to reduce proteinuria independent of ACE inhibitor and also reduce adverse cardiovascular outcomes (HR: 0.65, $P = .042$), but had no effect on adverse renal outcomes compared with placebo.¹⁷² Another post hoc analysis of this trial investigated the impact of systolic blood pressure on renal and cardiovascular outcomes.¹⁷³ A total of 566 Asian patients were followed for a mean of 3.2 years, and the results showed that in patients with proteinuria ≥ 1 g/gCr, a mean follow-up systolic blood pressure of >130 mm Hg was associated with an HR of 2.33 (1.62–3.36) for renal composite outcome comprising doubling of serum creatinine, ESKD, or death, compared with patients whose mean follow-up systolic blood pressure was ≤ 130 mm Hg. In patients without a history of cardiovascular disease, a follow-up mean systolic blood pressure of >140 mm Hg was associated with an HR of 2.04 (1.23–3.40) for cardiovascular outcomes compared with those whose systolic blood pressure was <140 mm Hg. The median slopes of eGFR were -3.27 , -4.53 , and -7.13 dL/mg/year in patients with systolic blood pressure ≤ 130 , 131–140, and >140 mm Hg, respectively ($P = .008$ between ≤ 130 and 131–140, $P < .001$ between ≤ 130 and >140 mm Hg). The investigators thus concluded that in Asian type 2 diabetic patients with CKD and heavy proteinuria, blood pressure control should be

individualized and a systolic blood pressure not >130 mm Hg was associated with renoprotection.¹⁷³

ARISTOLOCHIC ACID NEPHROPATHY

The story of AA nephropathy began in the early 1990s when a group of young females in Belgium was noted to develop tubulointerstitial nephritis with rapidly progressive deterioration of kidney function following the intake of slimming medications containing herbal ingredients.^{11,174} Subsequently the causative nephrotoxic agent was identified as AA,¹⁷⁵ present in the Chinese herb Guang Fang Ji, and poisoning occurred when this was mistaken as Han Fang Ji, a diuretic herb in traditional Chinese medicine.¹⁷⁶ It was also noted that AA exposure predisposed to the development of urothelial malignancies, so that AA was classified as a human carcinogen class I by the World Health Organization (WHO) International Agency for Research on Cancer in 2002.^{18,177,178} Kidney disease related to AA exposure is termed AAN. In the following years, cases of AAN have been reported all over the world, and areas with high prevalence rates include China, Taiwan, and countries in Eastern and Southeastern Europe near the Danube tributaries, where the entity of Balkan endemic nephropathy was recognized since the 1950s.^{179,180} After noting the similarities in renal histopathology and the high incidence of upper tract urothelial cancer between AAN and Balkan endemic nephropathy, the etiological link to environmental exposure to AA has been confirmed,^{181,182} and the evidence includes the detection of AA-DNA adducts and the hallmark A:T to T:A base transversion in renal cortical and urothelial malignant tissues.¹⁸³

Clinical Relevance

Aristolochic Acid Nephropathy

Pathogenesis in aristolochic acid nephropathy involves (i) injury to renal tubular cells with altered oxidative pathways and mitochondrial damage, followed by apoptosis, resulting in interstitial fibrosis and tubular atrophy; (ii) aristolactams activated by peroxidase at the urothelium, forming adducts with DNA, resulting in A:T to T:A base transversion in the TP53 tumor suppression gene, thereby affecting cell-cycle checkpoints, DNA repair, and apoptosis, leading to carcinogenesis.

AA is present in plants of the *Aristolochia* genus containing nitrophenanthrene carboxylic acids, of which 8-methoxy-6-nitro-phenanthro-(3,4-d)-1,3-dioxolo-5-carboxylic acid (aristolochic acid I, AAI) and 6-nitro-phenanthro-(3,4-d)-1,3-dioxolo-5-carboxylic acid (AAII) are the predominant components.¹⁸⁴ In Asia, exposure to AA is mostly related to intake of herbal remedies, whereas in patients with Balkan endemic nephropathy the data to date suggest that AA might have been ingested via bread made with flour contaminated by *Aristolochia clematitis*.^{185–187} AA has been identified in different Chinese medicinal herbs (Table 80.2).¹⁸⁶ Pathogenesis involves injury to renal tubular cells with altered oxidative pathways and mitochondrial damage, followed by apoptosis and impaired regenerative potential, resulting in interstitial fibrosis and tubular atrophy.^{188–190} Human urothelium is rich in peroxidases; the aristolactams activated by peroxidase then

Table 80.2 Chinese Medicinal Herbs That Contain Aristolochic Acids

Name of Chinese Herb	Botanical Name
Guang Fang Ji (Fangchi)	<i>Aristolochia fangchi</i>
Xixin	Radix et Rhizoma Asari
Guan Mu Tong	<i>Aristolochia manshuriensis</i>
Qing Mu Xiang	<i>Aristolochia cucurbitifolia</i>
Ma Dou Ling	<i>Aristolochia debilis</i>
Tian Xian Teng	<i>Aristolochia contorta</i>

Modified from Yang HY, Chen PC, Wang JD. Chinese herbs containing aristolochic acid associated with renal failure and urothelial carcinoma: a review from epidemiologic observations to causal inference. *Biomed Res Int*. 2014;2014:569325.

form adducts with DNA, which have been associated with A:T to T:A base transversion in the TP53 tumor suppression gene.^{184,187,191,192} TP53 is involved in cell-cycle checkpoints, DNA repair, and apoptosis,¹⁹³ thereby explaining the carcinogenic effect of AA. Characteristic AA-DNA adducts in the renal tissue such as 7-(deoxyadenosin-N6-yl) aristolactam I (dA-AAI) can serve as biomarkers of prior AA exposure.^{18,184,187,194} This deoxyadenosine adduct causes characteristic A-T transversion mutations and such mutations were found in the TP53 tumor suppressor gene in tumors from AAN and patients with Balkan endemic nephropathy.^{181,195} More recently, using whole-genome and exome sequencing analysis, A:T transversion mutations were also observed in other gene loci.^{196–198}

AAN is characterized by extensive cortical interstitial fibrosis with relative sparing of glomeruli and urinary tract urothelial carcinomas. Most cases presented with renal failure, moderate hypertension, and severe anemia, and approximately 70% showed rapid progression to ESKD.^{188,199} Some patients presented with Fanconi syndrome.^{200,201} Proteinuria was relatively mild due to its tubular origin.²⁰² Kidney biopsies showed extensive interstitial fibrosis and tubular atrophy, often starting from the peripheral cortex.^{11,174} Only in the late stage did the glomeruli show collapse of the capillaries and wrinkling of the glomerular basement membrane. Interstitial infiltrating cells were mostly not prominent, though inflammatory infiltration was observed in some renal biopsies, suggesting that immune-mediated injury could not be totally excluded.¹⁸⁰ While there is no pathognomonic diagnostic feature, the pattern of injury in the absence of other diseases would suggest the diagnosis of AAN. In a series of 300 Chinese patients with AAN, 13 patients presented with AKI, 7 patients presented with tubular dysfunction with relatively normal serum creatinine, and 280 patients presented with CKD.²⁰³ It is noteworthy that AKI was associated with a high level of AA exposure and they progressed to kidney failure over 1 to 7 years. By contrast, patients presenting with predominantly tubular dysfunction and relatively normal serum creatinine showed slow CKD progression rates. These observations suggested a dose-dependent relationship between AA exposure and the severity of renal injury.²⁰⁴ In this regard, cumulative exposure to 61–100 g of Mu Tong or Fangchi (herbs that contain AA) had been associated with increased risk of ESKD

with odds ratios of 1.47 and 1.60, respectively, and the odds ratios increased to 5.82 and 1.94, respectively, when the exposure was >200 g.²⁰⁵

The causative link between AA exposure and urothelial malignancies emerged from clinical and animal studies.^{17,18,191,206–210} Some patients first developed upper urinary tract urothelial carcinoma followed by subsequent bladder cancer, and it must be remembered that malignancies may develop at variable periods after the diagnosis of AAN, when patients are already on long-term dialysis or after kidney transplantation.^{18,211} The risk for malignancies is related to the cumulative dose of AA exposure, and cumulative exposure to >200 g of Guang Fang Ji was associated with increased risk of urothelial carcinoma.¹⁸ Prophylactic removal of native kidneys and ureters is recommended for some patients, especially those planned for kidney transplantation or with high-level prior exposure.¹⁸ The incidence of upper tract urothelial cancer appears particularly high in Eastern European patients diagnosed with Balkan endemic nephropathy and in patients with AAN in Taiwan, the latter especially in females.^{212,213}

Ultra-high-performance liquid chromatography–multistage fragmentation mass spectrometry can be used to detect AAI and AAI in herbal products,²¹⁴ and a hollow fiber liquid-phase microextraction technique in conjunction with high-performance liquid chromatography has been used for the extraction and quantitation of AAI in human urine.²¹⁵ AA-DNA adducts in the renal cortex or urothelial tissues can serve as biomarkers of prior AA exposure,^{18,184,194} using ³²P labeling or liquid chromatography–mass spectrometry, or ultraperformance liquid chromatography–triple quadrupole mass spectrometry of exfoliated urothelial cells.^{216–219}

Continued exposure to AA should be stopped. Patients with tubular dysfunction or CKD should be managed accordingly. Standard algorithms for diagnosis and management have been devised for subjects at increased risk of Balkan endemic nephropathy based on geographical location or family history.¹⁷⁹ Whether similar approaches could be applicable to Asian subjects with AAN remains to be investigated. In view of the inflammatory cell infiltration in the renal interstitium,²²⁰ the effect of corticosteroid therapy has been investigated. There are data in Caucasian AAN patients to suggest that corticosteroid therapy may reduce the rate of progression to ESKD.^{221,222} In a trial that included 43 Chinese patients with AAN, 25 patients were treated with prednisolone at 0.5 mg/kg body weight daily for 1–3 months, which was then reduced by 0.05 mg/kg every month and maintained at 10 mg daily, and the results suggested slower rate of CKD progression in the steroid-treated group.²²³ Subjects who have taken AA-containing herbal remedies, or those with occupational or environmental AA exposure, should have regular follow-up for the monitoring of kidney function and manifestations suggestive of uroepithelial malignancies. Patients with histopathologic features of endemic nephropathy and those in CKD stage 3 or above, including those on dialysis or after kidney transplantation, are considered at increased risk for developing upper tract urothelial cancer and are recommended to have urine cytology and imaging every 6 months according to the European consensus on endemic nephropathy.¹⁷⁹ This also applies to patients with previous bladder cancer. In view of the high incidence of urothelial carcinoma in patients with ESKD due to endemic nephropathy, prophylactic bilateral nephroureterectomy has been recommended with

determination of the level of AA-DNA adducts in the resected renal cortical tissue, especially in kidney transplant recipients before or after transplantation.^{179,224}

ANCA-ASSOCIATED VASCULITIDES

Antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV) comprises granulomatosis with polyangiitis (GPA), eosinophilic GPA, microscopic polyangiitis (MPA), and the renal-limited variant of MPA.²²⁵ ANCAs are serologic hallmarks of these small-vessel vasculitides. Proteinase 3 (PR3) and myeloperoxidase (MPO) are two major target antigens of ANCA in AAV. The histopathologic hallmark of ANCA-associated glomerulonephritis (ANCA-GN) is pauci-immune necrotizing glomerulonephritis, often with abundant crescents. While the typical finding is little or no glomerular staining for immunoglobulins by immunofluorescence microscopy, immune complexes could be detected in renal biopsies of some patients.²²⁶

Pathogenesis involves the interaction between genetic predisposition, environmental triggers such as infection, and effector mechanisms in the innate and adaptive immune system. Acute vascular inflammation is induced when resting neutrophils that have ANCA autoantigens sequestered in cytoplasmic granules are exposed to priming factors such as cytokines and proinflammatory mediators induced by infection or complement activation, triggering the appearance of ANCA antigens on the surface of neutrophils and their release into the surrounding microenvironment. ANCA then bind to these ANCA antigens on the primed neutrophils, resulting in neutrophil activation and the release of neutrophil extracellular traps.²²⁷ This and other mediators released by activated neutrophils activate the alternative complement pathway, which generates C5a, that acts as a chemoattractant for neutrophils and also primes the neutrophils for activation by ANCA. Activated neutrophils also adhere to and penetrate vessel walls and release oxygen radicals and destructive enzymes that induce local damage in the form of necrotizing inflammation of small vessels.²²⁸

AAV is a common autoimmune disease in Caucasians, with an annual incidence rate of 10–20 pmp and peak age of disease onset of 65–74 years.²²⁹ In Asia, AAV is an important glomerular disease in elderly patients presenting with AKI.²³⁰ In contrast to Caucasian populations, where GPA and anti-PR3 are more common, data from China and Japan showed a preponderance of MPA and anti-MPO disease, often associated with more chronic lesions in the kidney biopsy.^{231–236} MPA or GPA mostly occurs in older adults, although it has been reported at all ages. A series that included 234 Chinese patients with AAV showed that >40% of patients were above the age of 65 years at diagnosis and 94.9% of the elderly patients had anti-MPO antibodies.²³⁷ Data from Japan also showed that the annual incidence of MPA was approximately 10 times higher in subjects above the age of 65 years than in younger adults.²³⁵ In addition, compared with younger patients, older patients had more pulmonary involvement and a higher risk of infections. The typical presentation of ANCA-GN in Asia is rapidly progressive glomerulonephritis, often in an elderly patient. A retrospective study in China showed that 27 (12.1%) of 223 patients with ANCA-GN showed concomitant membranous nephropathy, and these patients were characterized by a higher level of proteinuria and more

severe renal injury and tubulointerstitial disease compared with those without membranous features in the kidney biopsy, and a relatively low seropositive rate for anti-PLA2R antibodies compared with patients with primary membranous nephropathy (12.5% vs. 65.0%).²³⁸

The 2015 European League Against Rheumatism and European Renal Association–European Dialysis and Transplant Association (EULAR/ERA-EDTA) recommendations for the management of AAV suggest that glucocorticoids combined with either cyclophosphamide or rituximab be used for remission-induction treatment of severe organ-threatening disease, with the additional consideration of plasma exchange in patients presenting with serum creatinine ≥ 500 $\mu\text{mol/L}$ due to rapidly progressive glomerulonephritis or those with alveolar hemorrhage, and that low-dose glucocorticoids combined with azathioprine, rituximab, methotrexate, or mycophenolate mofetil be used for remission maintenance.²³⁹ Corticosteroids combined with cyclophosphamide is still the most commonly used treatment for remission induction.^{240–242} In view of the predilection for elderly subjects and a high incidence of infective complications, treatment decisions with regard to the dose of immunosuppression entail a delicate balance between efficacy and potential treatment-related adverse effects. Data from a study in China that included 398 patients with AAV showed that 135 (33.9%) died during follow-up, with 83 deaths occurring within the first 12 months after diagnosis and infection being the leading cause of death (39 cases). Besides, older age, pulmonary involvement, and poor renal function at presentation predicted mortality.²⁴³ In patients presenting with rapidly progressive glomerulonephritis that required dialysis, infection (47.4%) and active vasculitis (26.3%) were the leading causes of death.²⁴⁴ The high incidence and poor outcome of infections in this patient population have also been reported from Japan, and the excessive mortality was attributed to cyclophosphamide, so that many patients are treated with corticosteroids alone and a starting dose of prednisolone <0.8 mg/kg per day has been recommended especially for elderly patients.^{245,246} In this context, data from a retrospective study in 82 Japanese patients with ANCA-positive MPA, 29 of whom were treated with corticosteroids plus cyclophosphamide while 53 were treated with corticosteroids alone, showed similar remission rates of >90% in both groups but a lower 5-year survival rate in those given cyclophosphamide (50% vs. 73% in the noncyclophosphamide group, $P = .041$).²⁴⁷ Another study reported that disease severity at presentation was associated with both the risk of relapse and infective complications during follow-up.²⁴⁸

The clinical outcomes including the survival of patients with AAV have improved substantially over the past decade. In a study of 398 patients >30% died after mean follow-up of approximately 2 years.²⁴³ Data from Japan showed improved renal and patient prognosis in patients presenting with ANCA-positive MPA with rapidly progressive glomerulonephritis in recent years, which was attributed to earlier diagnosis and improved treatment.^{246,249}

HEPATITIS B/C-ASSOCIATED RENAL DISEASES

Infection with HBV or hepatitis C virus (HCV) may be associated with a variety of glomerular diseases due to immune complex deposition or immune complex-mediated

vasculitis,^{250–253} with or without cryoglobulinemia,²⁵⁴ the latter being more common in the case of HCV infection.

HBV, a hepadnavirus with the virion consisting of a 42-nm spherical, double-shelled particle composed of small spheres and rods and with an average width of 22 nm, is partially double-stranded DNA. In endemic areas the transmission of HBV is mostly vertical from infected mother to child, followed by horizontal transmission within families or horizontal infection of adults who do not have protective immunity. In areas of intermediate prevalence transmission is mostly horizontal, affecting older children, adolescents, and adults. In areas of low prevalence, HBV is primarily a disease of adolescents and young adults, through sexual or parenteral acquisition.²⁵⁵ In the Far East prior to the advent of vaccination programs, vertical maternal–infant transmission of HBV during the perinatal period had been the predominant route of infection accounting for persistent endemicity.²⁵⁶ The age of presentation for HBV-associated kidney diseases thus varies according to the time of infection. Mass HBV vaccination programs markedly reduced the HBsAg carriage rate and the consequent HBV-associated liver complications including hepatocellular carcinoma.²⁵⁷ For example, the nationwide HBV vaccination program of Taiwan that was introduced in 1984 reduced the HBsAg carrier rate in children from 9.8% to <1% in 2 decades.²⁵⁸ The frequency of HBV-associated membranous nephropathy among children infected with HBV also decreased from 11.6% in the period 1974–1984 to 2.1% in 1994–2004, and to 0% in 2004–2009.²⁵⁹ A similar impact of immunization program on HBV-associated membranous nephropathy, reducing the incidence by approximately 90%, had previously been observed in South Africa.²⁶⁰

Extrahepatic manifestations such as glomerulonephritis, vasculitis, or reactive arthritis can occur with either acute or, more commonly, chronic HBV infection. Pathogenesis is related to the formation of immune complexes followed by complement activation. Renal diseases secondary to HBV can take the form of membranous glomerulonephritis, membranoproliferative glomerulonephritis, polyarteritis nodosa, or mesangial proliferative glomerulonephritis. The low molecular weight of HBeAg (3×10^5 Da) has been speculated to account for its ability to traverse the glomerular basement membrane and form subepithelial immune deposits. In a report from China that included 5935 kidney biopsies performed in 2010–2015, of which 1038 (17.5%) were diagnosed as secondary glomerular diseases, HBV-associated glomerulonephritis accounted for 9.2% of the secondary cases.²⁶¹ The most common pattern of renal involvement in renal biopsy is membranous nephropathy, followed by membranoproliferative glomerulonephritis, polyarteritis nodosa, IgAN, and focal segmental glomerulosclerosis, although the latter two categories might represent coincidental findings.²⁵¹ HBV-associated membranous nephropathy is more common in children and in many cases resolves spontaneously when seroconversion from HBeAg seropositivity to anti-HBe seropositive status occurs, but spontaneous resolution of HBV-related glomerulonephritis is uncommon in adults, and some patients show progressive renal deterioration over time.²⁶² Confirmation of HBV etiology is by the demonstration of HBV antigens in the kidney biopsy, although this may not be positive in all cases and positive staining needs to be interpreted in the context of appropriate histopathologic features.²⁶³

Much of the data on the treatment of HBV-associated glomerular diseases are derived from patients with membranous nephropathy. Treatment with interferon or nucleoside analogs has been reported to lead to a reduction of proteinuria in patients with HBV-related membranous nephropathy. Interferon treatment given for 4–12 months was associated with sustained remission of proteinuria in 20%–100% of patients, usually occurring within 6 months of clearance of HBeAg.²⁶²

The M-type phospholipase A2 receptor (PLA2R) on the surface of podocytes has been identified as the major autoantigen in primary membranous nephropathy, and autoantibodies directed against PLA2R are mostly of IgG4 subclass.^{264,265} Up to 80% of patients with primary membranous nephropathy, including patients in Asia, have circulating anti-PLA2R antibodies.^{266–271} Data from a recent study in China that included 179 patients with primary membranous nephropathy, 40 patients with membranous lupus nephritis, and 26 patients with HBV-associated membranous nephropathy showed glomerular PLA2R expression, in a fine granular pattern along the glomerular capillary wall, in 92.2% of patients with primary membranous nephropathy and 7.7% of patients with HBV-associated membranous nephropathy.²⁷² In addition, 93.3% of the primary membranous nephropathy cases showed prominent glomerular IgG4 deposition, while IgG3 was the predominant subclass in patients with HBV-associated membranous nephropathy, with IgG4 present in 11.5%.²⁷²

While more often associated with HCV than HBV,^{254,273–275} cryoglobulinemic renal disease has been reported in China, which has an HBV prevalence rate of 7.18% in the general population.²⁷⁶ A series of 12 patients diagnosed with HBV-associated cryoglobulinemia in 2008–2015 showed that renal involvement, characterized by nephrotic range proteinuria with microscopic hematuria, affected 75% of patients while 58.3% presented with skin rash.²⁷⁷ All patients showed reduced complement level and tested positive for rheumatoid factor. Kidney biopsies showed membranoproliferative glomerulonephritis, but HBV antigens were not demonstrable in all patients. All the patients were treated with glucocorticoids alone or in combination with other immunosuppressive agents, and 50% died or reached end-stage renal failure.²⁷⁷

The body of data to date suggests that antiviral therapy improves HBV-associated membranous nephropathy. However, this conclusion is based on a relatively small number of subjects and not from randomized controlled trials.^{278–280} A recent meta-analysis included 325 Asian patients with HBV-associated glomerulonephritis.²⁸¹ A total of 219 (215 Chinese and 4 Thai) patients had received treatment and were compared with 106 controls (96 Chinese and 10 Thai). The results showed an overall efficacy of nucleoside/tide in reducing proteinuria, resulting in a 3.6-fold increased rate of proteinuria remission.²⁸¹ Another meta-analysis that included 127 patients also yielded results supportive of treatment efficacy, applicable to both interferon and nucleoside/tide analogs.²⁸² The generalizability of these results to other patient populations is unclear.

Based on available data, and considering the safety of entecavir, which is a recommended standard of care treatment for HBV, antiviral therapy is recommended in patients with HBV-associated renal disease and evidence of active viral replication as indicated by the level of HBV DNA.^{283,284}

According to prevailing HBV treatment recommendations, nucleoside/tide analog treatment is continued for at least 12 months after seroconversion of HBeAg (to anti-HBe), whereas patients who are HBeAg negative at the start of treatment often require lifelong therapy.²⁸⁵ In patients with HBV-associated renal disease, immunosuppression should be reserved to those with rapidly progressive glomerulonephritis or severe polyarteritis nodosa, with plasmapheresis considered in cryoglobulinemic cases.

Recent epidemiological reports showed a prevalence rate of approximately 3%–4% for HCV infection in the general population across Asia, varying from 0.08% in Hong Kong, 0.37% in Singapore, 0.47% in the Philippines, 0.49% in Japan, 0.6%–1.3% in Korea, 1.6%–3.2% in China, 2.2% in Thailand, to 1.8%–4.4% in Taiwan.^{286–289} The predominant genotypes also vary, with 1b accounting for around 70% in China and around 40% in Korea, whereas genotype 3 accounts for up to 80% of cases in India and Pakistan, and genotype 6 restricted to a few places such as Hong Kong and Thailand.²⁸⁷ Prevalence is higher with increasing age, among injecting drug users (48%–90%), and those with coinfection by human immunodeficiency virus (32%–85%). Annual incidence of HCV in China was estimated as 6.01/100,000.²⁸⁸

Extrahepatic manifestations occur in 38%–76% of patients with chronic HCV infection, mostly as immune-mediated diseases such as mixed cryoglobulinemia, Sjögren syndrome, autoimmune thyroiditis, and B-cell lymphoproliferative disorders.²⁹⁰ Renal involvement, mostly as membranoproliferative glomerulonephritis, occurs in approximately 20% of patients with mixed cryoglobulinemia.²⁹¹ For extrahepatic manifestations of HCV, including glomerulonephritis and cryoglobulinemic vasculitic syndromes, treatment with direct-acting antiviral agents has demonstrated efficacy and is recommended.^{291,292} Antiviral efficacy and improvement of renal abnormalities have been demonstrated in Asian patients with HCV-associated cryoglobulinemic membranoproliferative glomerulonephritis.^{293,294} However, a recent case report from the United States described a patient with recurrent mixed cryoglobulinemia syndrome despite sustained virologic response after treatment with direct-acting antiviral agent.²⁹⁵ In addition, the high cost of direct-acting antiviral agents presents a substantial financial barrier to treatment access in many Asian countries, as is the case with rituximab for the treatment of HCV-associated cryoglobulinemic vasculitis.

KIDNEY INJURY RELATED TO INFECTIONS

LEPTOSPIROSIS

Leptospirosis is the most widespread zoonosis and is endemic worldwide, being most prevalent in tropical, humid, and rural environments, particularly in warm and humid regions.²⁹⁶ The disease is caused by the spirochete *Leptospira*, which enters the human body through direct cutaneous or mucosal transmission or by aerosolization.²⁹⁷ Rodents and small mammals are reservoirs for *Leptospira*, and infected animals shed the spirochete in their urine into water or soil, which then infect humans through occupational (as occurs in farmers, sewage workers, and miners), household environmental, or recreational exposure. Increased risk of exposure to *Leptospira* after flooding has been observed.²⁹⁸ Each

Leptospira serovar is frequently associated with a particular maintenance host. Results of genotype investigation using multiple-locus variable-number tandem repeat analysis, including 11 loci in 110 *Leptospira interrogans* serogroups from Japan, the Philippines, Taiwan, and Vietnam, and 4 loci in 52 *Leptospira borgpetersenii* isolates from Japan, the Philippines, and Taiwan, showed variable genetic diversity in different serogroup strains and a wider host range for *L. borgpetersenii* compared with *L. interrogans*.²⁹⁹

Approximately 90% of *Leptospira* infections are subclinical or associated with mild self-limited disease. A study in India that included serum samples from 244 miners detected antibodies to leptospires in 38.5% of subjects, compared with 13% in nonminer controls, and serogroup analysis showed concordance with carrier animals in the region.³⁰⁰ Leptospirosis is an important topic in travel medicine because the disease can present as an imported case in Western countries in travelers returning from Asia.³⁰¹ However, it can also occur in subjects with no recent travel history and in urban settings,³⁰² especially in areas with a large population of rodents. For example, it has been reported that 65% of live-trapped rats in Baltimore (USA) had antibodies to *L. interrogans*.³⁰³ Leptospirosis occurring in an endemic setting or presenting as epidemic outbreaks, especially in developing areas and crowded urban areas, occurs in different parts of the world including Thailand, the Philippines, and China.^{304–307} It has been recognized as a reemerging infectious disease in Japan,³⁰⁸ and a high prevalence rate of up to 50% of pathogenic *Leptospira* has been detected in wild and domesticated animals including rats and dogs in an endemic area in Jiangxi Province of China.³⁰⁹

Symptoms develop after an incubation period of 5 to 14 days. Mild disease has nonspecific symptoms that are indistinguishable from other febrile illnesses. Severe and potentially fatal disease characterized by fever; headache; myalgia; AKI due to acute interstitial nephritis, jaundice, and hemorrhagic complications; and multiorgan failure occurs in up to 10% of patients, and is associated with a mortality rate of 5% to 15%.²⁹⁷ Typical clinical manifestations of leptospirosis fall into four categories: 1. mild influenza-like illness; 2. Weil syndrome characterized by jaundice, renal failure, hemorrhage, and myocarditis with arrhythmias; 3. meningitis/meningoencephalitis; and 4. pulmonary hemorrhage with respiratory failure.³¹⁰ Infection with icterohemorrhagiae serogroups has been reported to be associated with an increased risk of severe disease or death.^{296,311} Cardinal features of severe leptospirosis include nonoliguric AKI, marked hyperbilirubinemia of >30 mg/dL (512 μmol/L) with less severe elevation of aminotransferase levels, thrombocytopenia, and lung involvement.²⁹⁶ AKI as a result of acute tubulointerstitial nephritis is an early manifestation.³¹² Oliguria, cardiac arrhythmia, and pulmonary involvement portend a poor prognosis.³¹³ Urinary sodium and potassium loss due to proximal renal tubular dysfunction resulting in electrolyte abnormalities is common.³¹⁴ Proximal tubular dysfunction such as bicarbonaturia, glycosuria, decreased proximal sodium reabsorption, phosphaturia, magnesuria, and uricosuria can occur.³¹⁵ Rhabdomyolysis can also contribute to the AKI.³¹⁶ The manifestation of severe disease can be variable. In some patients after the initial septicemic phase there is a period of transient symptomatic improvement before an immune phase in which critical disease manifestations

can occur, whereas in other patients there is progressive symptomatic deterioration to fulminant disease.²⁹⁶ The triad of fever, jaundice, and AKI in an acutely ill patient should alert the clinician of possible leptospirosis.

Leptospira reach the kidneys via hematogenous spread and finally localize in the proximal tubular lumen.³¹⁷ Guinea pigs infected with *Leptospira* showed renal tubular cell injury, interstitial nephritis, and associated microvascular injury, particularly at the corticomedullary junction.³¹⁸ Ultrastructural studies in mice showed the entry of *Leptospira* from the capillary lumen to the renal interstitial tissue inducing interstitial edema and cellular infiltration after 4–8 days.³¹⁹ Leptospiral antigens are found in the proximal tubule cells and infiltrating macrophages, and also in the form of large extracellular clumps in the interstitium.³²⁰ Expression of leptospiral glycolipoprotein in renal tubules and vascular lumen has been implicated in the pathogenesis of acute tubulointerstitial nephritis and chronic tubulointerstitial lesions in chronically infected rats.^{318,321} *Leptospira* outer membrane proteins (OMPs) may elicit tubular injury and inflammation through Toll-like receptor (TLR)-dependent pathways involving activation of nuclear transcription factor kappa B and mitogen-activated protein kinases, with downstream induction of chemokines and cytokines relevant to tubular inflammation.³¹² LipL32, a 32-kDa lipoprotein, is a prominent leptospiral protein that is highly conserved among pathogenic *Leptospira* species.³²² LipL32 has been shown to induce inflammatory cytokines and stimulate extracellular matrix production in cultured renal tubular epithelial cells through transforming growth factor- β 1 (TGF- β 1)/Smad-dependent pathway,^{323,324} and these proinflammatory steps involved Toll-like receptor 2 (TLR2) but not TLR4 in cultured renal tubular cells.^{325–327} LipL32 also recognized and adhered to components of extracellular matrix including laminin, collagen I, and collagen V.^{328,329} *Leptospira santarosai* serovar Shermani OMP increased the secretion of active TGF- β 1 by proximal renal tubular epithelial cells and increased nuclear translocation of Smad3, accompanied by increased type I and type IV collagen messenger RNA and protein expression.³²⁴ In chronic infection *Leptospira* may colonize and persist in the proximal renal tubule, and this has been associated with chronic tubulointerstitial nephritis and fibrosis.³³⁰ Results from a recent epidemiological study showed an association between previous *Leptospira* exposure and impaired kidney function, suggesting that previous or chronic leptospiral infection is a risk factor for CKD.³³¹

The diagnosis of leptospirosis depends mainly on serologic methods such as microscopic agglutination test to detect antibody to *Leptospira*, polymerase chain reaction (PCR) to detect *Leptospira* DNA, or culture of the microorganism.²⁹⁷ The diagnosis of leptospirosis can be challenging to confirm, as the organism requires specialized culture media and grows over a period of weeks.^{296,297} Serologic testing is often negative early in the course of the disease,³³² and suboptimal sensitivity and specificity rates have been reported with an IgM enzyme-linked immunosorbent assay (ELISA) for the diagnosis of acute leptospirosis.³³³ PCR-based nucleic acid amplification testing of blood, urine, or cerebrospinal fluid is much more sensitive than culturing and can be performed early in the course of the disease, but such testing is not widely available.³⁰² Given these limitations, most diagnoses are still confirmed by serologic testing. A single indirect hemagglutination titer

of at least 1:200 but <1:800 is suggestive of *Leptospira* infection, and infection is confirmed by either a single titer of $\geq$ 1:800 or by a fourfold or higher increase in titer in the convalescent serum specimen.³³⁴

Guidelines and expert opinion support prompt treatment with antibiotics in suspected and confirmed cases.^{310,313} A review of seven randomized trials showed insufficient evidence for or against antibiotic therapy, but the duration of disease appeared shorter in patients treated with antibiotics though the difference did not reach statistical significance.³³⁵ Clinical studies have demonstrated the effectiveness of early antibiotic treatment for leptospirosis.³³⁶ It has been shown that serum creatinine level improved after 2.7 ± 2.2 days of IV penicillin therapy.³³⁷ Treatment with effective antibiotics should be initiated as soon as leptospirosis is suspected based on clinical features before microbiological confirmation. Delayed initiation of antibiotics by $\geq$ 2 days was associated with more severe disease.³¹¹ While a trial comparing ceftriaxone (1 g daily) with penicillin (1.5 million units every 6 hours) for 7 days showed no significant difference in the time to resolution of fever, in general IV penicillin is used to treat severe disease and oral doxycycline is used to treat mild disease.³³⁸ Adults with severe disease should receive either penicillin (1.5 million units IV every 6 hours), doxycycline (100 mg IV twice daily), ceftriaxone (1–2 g IV once daily), or cefotaxime (1 g IV every 6 hours) for a duration of at least 7 days. Pregnant women with severe leptospirosis may be treated with IV penicillin, ceftriaxone, cefotaxime, or azithromycin, but not doxycycline. Patients with mild disease can be treated with doxycycline (adults: 100 mg orally twice daily) or azithromycin (adults: 500 mg orally once daily for 3 days), and pregnant women should be treated with either azithromycin or amoxicillin. Jarisch–Herxheimer reactions may occur following effective treatment. In addition, it is advisable to follow the kidney function of these patients in view of the recent data suggesting an association between prior leptospirosis and CKD.³³¹

Clinical Relevance Leptospirosis

Localization of leptospiral antigens and glycolipoprotein in the proximal tubular cells, infiltrating macrophages, and vascular lumen, or in the form of extracellular clumps in the interstitium, are implicated in the pathogenesis of acute tubulointerstitial nephritis and chronic tubulointerstitial lesions. *Leptospira* outer membrane proteins such as LipL32 may elicit tubular injury and inflammation through pathways dependent on Toll-like receptors and involving the activation of nuclear transcription factor kappa B and mitogen-activated protein kinases, with downstream induction of chemokines and cytokines such as the transforming growth factor- β 1 (TGF- β 1)/Smad-dependent pathway that are relevant to inflammation and fibrosis. Chronic *Leptospira* colonization in the proximal renal tubule is associated with chronic tubulointerstitial nephritis and fibrosis. These findings, together with data from a recent epidemiologic study in Taiwan, suggest potential contribution of previous or persistent *Leptospira* exposure to chronic kidney disease.

DENGUE VIRAL INFECTION

Dengue is a common arthropod-borne viral (arbovirus) disease in tropical and subtropical regions.^{339–341} While the annual incidence is estimated as 50 to 200 million by the WHO,³⁴² this could be an underestimate because the symptoms and signs are nonspecific, with fever being one of the most common clinical manifestations.³⁴³ It is endemic in >100 countries, and an estimated 2.5 billion of the world's population are at risk of dengue infection.^{344,345} The incidence of dengue fever has increased 30-fold in the past 50 years, and factors such as global warming, increased air travel and globalization, urbanization, and ineffective mosquito control have resulted in the spread of Dengue virus and the introduction of multiple viral serotypes into permissive areas, resulting in most tropical regions becoming hyperendemic with multiple viral serotypes cocirculating (Fig. 80.5).^{346–352} The disease is common in South and Southeast Asia in countries such as Malaysia, Thailand, Indonesia, Vietnam, and Singapore,^{339,353–355} and is spreading to other parts of Asia such as Japan, Korea, China, Taiwan, and Hong Kong.^{343,347–351,356}

Dengue is transmitted by *Aedes* mosquitoes, principally *Aedes aegypti*, and is caused by four viral serotypes of the genus *Flavivirus* (DENV-1 to DENV-4).³⁴⁴ Infection with one

serotype confers lifelong immunity to that type but only short-term immunity to the others, and subsequent infection with a different serotype is associated with increased risk of severe complications. The female mosquito becomes infected when she takes a blood meal from an infected subject during the acute febrile and viremic phase of illness. During the extrinsic incubation period, the virus first infects midgut cells and then disseminates to replicate in numerous mosquito tissues, ultimately infecting the salivary glands after 8–10 days. Once the salivary glands are infected, the mosquito is infective and can transmit the virus to another person. The mosquito remains infective for life. There is no specific antimicrobial therapy. DENV vaccines are currently under development, including some in phase III safety and efficacy testing. Prevention is primarily by reducing the habitat and the number of mosquitoes, and limiting exposure to bites.

The time from infection to onset of illness, termed the intrinsic incubation period in humans, ranges from 3 to 14 days, with an average of 4 to 7 days. The symptoms that may appear include fever (often the first manifestation and can be the only manifestation in children), headache, myalgia, joint pain, and a measles-like skin rash which lasts up to 7 days. Slight abdominal pain and diarrhea can occur. Leukocyte

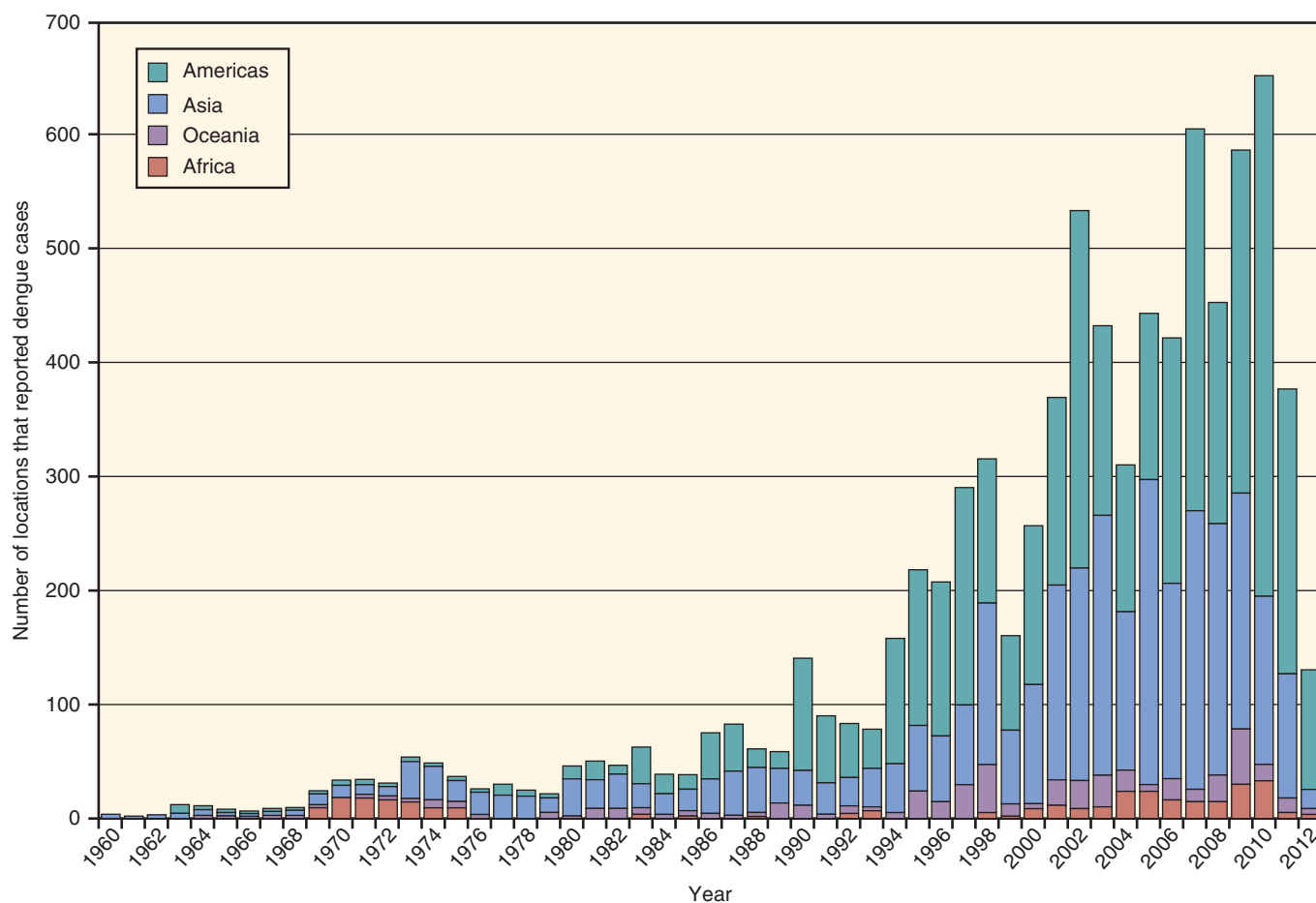


Fig. 80.5 Changing epidemiology of dengue over time. (Reprinted by permission from Macmillan Publishers Ltd. Guzman MG, Gubler DJ, Izquierdo A, et al. Dengue infection. *Nat Rev Dis Primers* 2016;2:16055; adapted from Messina JP, Brady OJ, Pigott DM, et al. A global compendium of human dengue virus occurrence. *Sci Data* 2014;1:140004.).

counts are usually decreased. Patients can become better or deteriorate after 2–5 days. A small proportion of patients develop life-threatening dengue hemorrhagic fever (DHF), with features of thrombocytopenia, petechiae or ecchymosis, disseminated intravascular coagulation, and plasma leak resulting in hemoconcentration, hypoproteinemia, pleural effusion, and ascites. More severe cases are termed dengue shock syndrome (DSS), in which systemic vascular leak and severe hypotension occur.³⁴³ There can be concomitant hepatitis, neurologic disorders, and myocarditis. Mortality rate in untreated cases could reach 20%, while supportive treatment including IV hydration can reduce the mortality rate to <1%.³⁴⁴ In 2009, the WHO revised the classification as dengue with or without warning signs, and severe dengue, with the objective of facilitating the identification of patients at risk of severe disease.³⁴⁶ It has been suggested that severe manifestations such as DHF and DSS occur as a result of antibody-dependent enhancement when a dengue-immune subject acquires infection by a different DENV serotype.³⁵⁷ The facilitation of DENV lineage dispersal by air traffic of humans and/or mosquitoes thus has potential clinical and public health implications, and recent investigations showed that large transportation hubs such as Thailand could act as central points of DENV lineage movement in Asia.³⁵²

Proteinuria is common in severe dengue, and a detection rate of 74% has been reported.³⁵⁸ The underlying pathogenesis of renal abnormalities in dengue is probably multifactorial. AKI is an infrequent complication of dengue infection, occurring in up to 13% of patients depending on the severity of infection, but is associated with increased morbidity and mortality, and a mortality rate of 64% has been reported.^{359–362} In severe dengue infection hemodynamic disturbance, acute hemolysis, and rhabdomyolysis may contribute to direct kidney injury.^{363–366} Immunologic mechanisms might also play a role, as suggested by renal abnormalities in the absence of shock, the demonstration of immune complex-mediated mesangial proliferation in human and animal models, and association with reduced levels of complement components.^{367,368}

Anti-DENV IgM antibody detection with ELISA is most commonly used for diagnosis of acute disease, while patients in convalescence are identified through IgM and IgG seroconversion.³⁴⁶ For patients suspected of dengue, a presumptive diagnosis can be made by the detection of IgM antibodies in samples collected at day 6 of acute symptoms. DENV viremia is detectable 24–48 hours before fever onset and continues for 5–6 days. During this period, infective virus, its specific RNA, and the NS1 protein can be detected in patient blood, serum, and plasma, and also in tissues from fatal cases. If the acute-phase sample obtained within the first 3 days after onset of symptoms is negative for IgM antibodies, testing for dengue viral RNA or NS1 antigen can be performed. DENV RNA detection provides a rapid, sensitive, and specific method for virological diagnostic confirmation. NS1 protein detection provides a window of opportunity for early etiological diagnosis. The sensitivity and specificity of DENV NS1 detection depend on the infecting serotype, the timing of sample collection, and the parity of DENV infection. Antibody cross-reaction is observed between DENV and Zika virus because both are Flaviviridae. Specific detection of DENV, chikungunya, and Zika viral RNA can be done by real-time PCR in acute samples. Under appropriate circumstances, IgM serology on serum samples collected from individuals with clinically

suspected DENV infection or Zika virus infection should be performed by IgM Capture ELISA for both DENV and Zika virus. If positive to both viruses, a secondary *Flavivirus* infection should be considered.

Supportive care and careful monitoring are essential, as there is no specific treatment for dengue. Treatment with corticosteroid or chloroquine was shown to be not beneficial.^{369,370} Fluid intake during the 24 hours before being seen by a clinician has been significantly associated with decreased risk for hospitalization.³⁷¹ Analgesic and antipyretic drugs such as paracetamol can be prescribed at the usual dosage for children and adults. Neither aspirin nor nonsteroidal antiinflammatory drugs should be taken. Vector control to avoid mosquito bites should be used to prevent disease transmission.

Fluid therapy is key to dengue management and is applied based on disease severity, and fluid lost due to capillary leakage must be replaced with IV crystalloid solutions (volume-expanding fluid replacements such as lactated Ringer's solution) or physiological (normal) saline solution. The subsidence of fever could portend clinical deterioration, and thus patients need to be carefully monitored.

MALARIA

Malaria is a protozoan disease caused by four main *Plasmodium* species (*Plasmodium falciparum*, *Plasmodium vivax*, *Plasmodium malariae*, and *Plasmodium ovale*) and the zoonotic parasite *Plasmodium knowlesi* found in many Southeast Asian countries.^{372,373} Over 200 million cases of malaria occurred in 2015 and the number of deaths exceeded 420,000.³⁷² The majority of infections occur in Africa (90%), followed by South/Southeast Asia (6%), and the eastern Mediterranean region (2%). Morbidity and mortality due to malaria have decreased globally secondary to malaria control programs over the last few decades, while a high level of transmission continues in some regions. In addition, the emergence of artemisinin-resistant *P. falciparum* malaria in Southeast Asia and the lack of a highly protective vaccine present challenges to continued progress.³⁷⁴

Malaria is transmitted by *Anopheles* mosquito. Infection in an individual without prior exposure almost invariably produces a febrile illness. The accompanying symptoms are nonspecific and often include rigors, headache, nausea, and muscle pains. If treated with appropriate drugs at this stage the symptoms remit over a few days, though often with considerable exhaustion. In the case of *P. falciparum*, complete treatment will eradicate the infection, and any return of symptoms reflects incomplete treatment, resistance to the drugs used, or new infection. In the case of *P. vivax* and *P. ovale*, subsequent infections may recur at intervals as a result of reactivation of the dormant liver-resident hypnozoite stage, unless this is cleared by a prolonged treatment with an 8-aminoquinoline drug.^{375–377} Clinical and laboratory criteria are used to establish the diagnosis of severe malaria, which requires aggressive treatment especially for travelers who have no prior immunity.^{376,378} The clinical features of malaria can simulate other febrile illnesses. Therefore the WHO recommends that diagnosis be established with a parasite-specific laboratory test in patients suspected to have malaria based on clinical grounds.³⁷⁶ Laboratory testing options vary based on the geographic, clinical, and socioeconomic setting.

The most commonly used methods for laboratory diagnosis include microscopic examination of stained blood films and detection of parasite antigen or nucleic acid,³⁷⁹ of which microscopic examination of thick and thin blood films remains the gold standard. Rapid antigen detection methods and molecular amplification tests are used increasingly as adjunctive tests. Tests for antiparasitoid antibodies are commercially available but are not recommended for diagnosis of acute disease.

About 120 types of *Plasmodium* species have been reported, although only five are accepted as human malaria parasites. Notably, *P. falciparum* and *P. vivax* cause most of the human infections worldwide. The most virulent species *P. falciparum* is also the most prevalent parasite in Africa, whereas *P. vivax* is the most widely distributed parasite outside of Africa, for example, in India.^{380,381} *P. malariae* and *P. ovale* spp are much less prevalent. *P. malariae* is more or less sympatric with *P. falciparum* in distribution, mainly found in sub-Saharan Africa and southwest Pacific.³⁸² *P. ovale* spp were thought to have a limited distribution, mostly found in Africa and some Pacific Islands.³⁸³ However, the prevalence of these two species is probably underestimated, due to low parasitemia, morphological resemblance with *P. vivax*, and occurrence as mixed infections.³⁸⁴ *P. malariae*, *P. ovale*, and *P. vivax* may cause febrile attacks years after exposure.³⁸⁵ Delayed clinical disease due to *P. ovale* or *P. vivax* is due to the existence of hypnozoites, a minority of parasites that remain in the liver instead of inducing a blood stage like the majority. These dormant microorganisms can remain viable for many years, thereby causing late onset of disease. For *P. malariae* a low level of parasitemia may persist for years, sometimes below the level of detection on a thick film, and disease can emerge more than a decade after primary infection.³⁸² Chronic *P. malariae* infection can result in immune complex formation, leading to membranoproliferative or mesangioproliferative glomerulonephritis and proteinuria (quartan malarial nephropathy).^{386,387} Molecular testing and genotyping have yielded more accurate data on distribution.³⁸⁸ In the Greater Mekong Subregion of Southeast Asia, including the countries of Cambodia, China, Laos, Myanmar, Thailand, and Vietnam, where there is a plan for regional malaria elimination by 2030, all five *Plasmodium* parasites infecting humans coexist and *P. vivax* and *P. falciparum* are the predominant parasite species.³⁸⁹ A recent study of 2701 blood samples collected from the China–Myanmar border region using nested PCR to detect the protozoan genome identified 561 malaria cases, including 161 *P. falciparum*, 327 *P. vivax*, 66 *P. falciparum*–*P. vivax* mixed infections, 4 *P. malariae*, and 3 *P. ovale* spp. *P. vivax* and *P. falciparum* accounted for >60% and approximately 30% of all malaria cases, respectively.³⁹⁰ In addition to South Asian countries, *P. malariae* infections have been described in China,³⁹¹ Thailand,^{392,393} and Vietnam,³⁹⁴ and may exhibit high genetic diversity.³⁹⁵ *P. ovale* spp have been found in South Asian and neighboring countries including Cambodia, Laos, Myanmar, Thailand, and Vietnam.³⁹² Most *P. ovale* spp reported in China have been imported, mainly from Africa.³⁹⁶ Indigenous malaria cases decreased significantly after the National Malaria Elimination Action Plan was initiated in China in 2010.³⁹⁷ A recent report from Jiangsu province in China showed that of the 1268 cases of malaria reported in the province in 2011–2014, 83.4% were imported *P. falciparum* cases, while 13.0% were due to *P. ovale*, and 2.8% were due

to *P. malariae*, many of which were imported from countries in Africa and South Asia.³⁹⁸

In Korea, *P. falciparum* infection was once present among IV drug abusers, while indigenous falciparum malaria has not been reported since 1945.³⁹⁹ *P. malariae* was also present before 1945, but has not occurred since. Vivax malaria had been prevalent in Korea for many centuries, but its incidence decreased rapidly from the 1970s. The Republic of Korea (South Korea) was once declared as malaria free in the late 1970s, but vivax malaria reemerged in 1993. Initially localized to the area bordering North Korea it has subsequently spread from west to east.^{400–402}

In the nonendemic setting of Singapore, a retrospective analysis of 214 patients with smear-positive malaria treated at a tertiary hospital in 2000–2010 showed that *P. vivax* accounted for 59.3% of cases, whereas *P. falciparum*, *P. malariae*, and mixed infections accounted for 38.8%, 0.3%, and 1.4% of cases, respectively.⁴⁰³ Most patients presented with fever and thrombocytopenia was common. The 43 severe cases were characterized by older patient age and higher level of parasitemia. All the patients survived.

P. knowlesi has emerged as the most common cause of malaria infection in Malaysia.^{404,405} Owing to its 24-hour erythrocytic stage, *P. knowlesi* replicates quickly.^{406,407} It can cause severe and fatal disease and thus it is important to diagnose and treat early.^{408–411} Knowlesi malaria shows a fatality rate that is comparable with, if not higher than, that of falciparum malaria.^{409,410} The severity of knowlesi malaria, including the complication of AKI, is significantly associated with high parasitemia level.^{410,412,413} Diagnosis with blood film microscopy may be associated with misidentification because the *P. knowlesi* ring stages resemble *P. falciparum* and later trophozoites resemble *P. malariae*. Newer methodologies such as loop-mediated isothermal amplification are being investigated.⁴¹⁴

Previous reports from India showed that the clinical presentation of severe falciparum malaria has more complications, including multiorgan failure often including AKI, which was associated with a higher mortality rate.^{415–417} Disease severity is usually lower with vivax malaria compared with falciparum malaria. However, there are marked variations and there have been increasing reports of severe vivax malaria, including respiratory distress,⁴¹⁸ AKI,⁴¹⁹ and cerebral malaria,⁴²⁰ and also reports of disease occurring in temperate regions.⁴²¹ In a series that included 341 Korean patients diagnosed in 2005–2009, including six recurrent cases, the common manifestations included fever (98.8%), thrombocytopenia (99.7%), chills (62.5%), transaminasemia (54.8%), leukopenia (43.4%), headache (33.7%), anemia (28.5%), and myalgia (24.9%). Less common but severe complications included hypotension (4.1%), altered mentation (0.9%), AKI (0.9%), spleen infarction (0.6%), and spleen rupture (0.3%).⁴²¹ Another series that included 210 Korean patients diagnosed with vivax malaria in 2006–2012 showed that 11 (5.2%) required intensive care unit admission and among these patients five required mechanical ventilation and one was treated with extracorporeal membrane oxygenation.³⁹⁹ Overall, the disease was categorized as severe in 21.0%, with pulmonary complications being most common (21.9%), followed by cerebral (2.4%), shock (1.9%), and bleeding complications (1.4%), and AKI occurred in 1.0% of patients. None of the patients died.

Pathogenesis leading to organ injury in severe malaria involves a complex interplay between direct damage due to parasite sequestration in microcirculation and host response with the release of various cytokines.^{422–425} Parasitized red cells adhere to vascular endothelial cell surface markers such as ICAM1.⁴²⁶ In this regard, cerebral malaria and multiorgan dysfunction (including AKI) have been associated with distinct profiles of cytokines and chemokines involving IL-17, IL-10, and human interferon-inducible protein 10 (IP-10).⁴²⁷ The incidence of AKI can be up to 60%.⁴²⁸ AKI complicating *P. falciparum* seems to have increased over the past decade.^{429–431} In addition, AKI has been associated with vivax malaria.^{419,432} AKI complicating *P. falciparum* malaria is associated with acute tubular necrosis with renal tubular cells showing variable degrees of cloudy swelling, degeneration, and necrosis. A recent report from Thailand showed predilection of injury at the distal tubule and a correlation between the degree of tubular injury and serum creatinine level. Changes in the renal tubules included cell vacuolation and degeneration, necrosis of individual tubular cells or tubular segments, scattered or diffused detachment of the renal tubular cells from the basement membrane, deposition of hemosiderin pigments, and minimal chronic inflammatory cell infiltration.⁴³³

SCRUB TYPHUS

Scrub typhus is a mite-borne infectious disease caused by the intracellular gram-negative bacteria *Orientia tsutsugamushi* (former name *Rickettsia tsutsugamushi*). Rodents are the main reservoir, and their mites act as both the reservoir and the vector. Human infection occurs when the larvae of the trombiculid mite infected with *O. tsutsugamushi* bite to suck human tissue fluid.^{434,435} An estimated 1 million cases occur each year,⁴³⁶ and the disease is common in Southeast Asia, Japan, China, and South Korea.^{437,438} Disease manifestations vary from mild or asymptomatic infection to severe potentially fatal illness, with clinical features of high fever, skin rash, headache, myalgia, lymphadenopathy, gastrointestinal symptoms, and cough. Most patients have a relatively benign course. A typical eschar occurs at the site of the bite, but is not always present. According to organ involvement, patients could present with pneumonitis, meningitis, encephalitis, myocarditis, acute pulmonary edema, pericarditis, hepatitis, or multiorgan failure.⁴³⁹ Severe disease is characterized by acute respiratory distress syndrome, AKI, bleeding, coagulation disorders, meningoencephalitis, and shock.^{440–444} Respiratory distress and encephalitis are the major causes of death.⁴⁴⁵ Renal involvement is common. Urinary abnormalities, such as hematuria, proteinuria, pyuria, and granular casts, occur in up to 80% of patients, and AKI occurs in 8%–60%.^{441–444,446–450} Pathogenic mechanisms include direct renal tubular cell injury with or without tubulointerstitial inflammatory cell infiltration.^{451,452} A higher serum level of neutrophil gelatinase-associated lipocalin has been shown to be associated with the development of AKI in scrub typhus.⁴⁵³ Intravascular hemolysis in patients with glucose-6-phosphate dehydrogenase deficiency, rhabdomyolysis, hypoperfusion, and vasculitis are other contributing factors to kidney injury.^{446,454} Histopathology shows acute tubular necrosis, interstitial nephritis, and mesangial proliferation.^{451,455–457} A retrospective study of 510 Korean patients with scrub typhus diagnosed in 2001–2013

showed that 35.9% developed AKI, and that risk factors for AKI included older age and concomitant comorbidities such as diabetes mellitus and preexisting CKDs.⁴⁵⁸ Furthermore, the complication of AKI was associated with higher rates of intensive care unit admission and death. All except one patient with AKI recovered. Serologic diagnosis of acute scrub typhus infection is based on a fourfold or higher increase in titer in paired samples drawn at least 14 days apart.⁴⁵⁹ Serum or eschar PCR to detect bacterial genome may also be used for diagnosis.^{460,461} Prompt treatment with doxycycline results in rapid improvement, including that of renal function.^{442,455} Elderly patients are at risk of more complications and poor clinical outcome, which could be related also to delayed presentation.⁴⁶²

SEVERE FEVER WITH THROMBOCYTOPENIA SYNDROME

Severe fever with thrombocytopenia syndrome (SFTS) was initially recognized in the Henan and Hubei provinces of China around 2007,⁴⁶³ and was subsequently reported in other provinces in China such as Zhejiang and Liaoning,^{464,465} South Korea,⁴⁶⁶ and Japan.⁴⁶⁷ The syndrome is caused by the SFTS virus (SFTSV; genus *Phlebovirus*, family *Bunyaviridae*), and ticks (e.g., *Rhipicephalus microplus*) in domestic animals are the vector.^{463,468–470} Investigations showed that SFTSV was common throughout different parts of Korea and China, especially in the warmer areas where ticks were more prevalent, and the disease mostly occurred in the months between April and October with farmers at higher risk due to occupational exposure.^{470–473} Patients infected with SFTSV presented with fever, thrombocytopenia, leukocytopenia, hemorrhagic features, and in severe cases multiorgan failure. Mortality rate was reported as 8.7%–47.2%.^{472,474} AKI occurred in about 20% of cases and was associated with a higher mortality rate of 12%–30%.⁴⁶³ Studies from China reported hemorrhagic features in >70% of patients, including gastrointestinal bleeding in 10% of patients, liver dysfunction in the majority, AKI requiring dialysis in 23%, and a 27% mortality rate.^{475,476} In a study that included 129 cases from Zhejiang province, China, the death rate was 12.4% and extremes of body mass index and delayed diagnosis were associated with increased risk of death.⁴⁷⁷ Clinical features of SFTS are nonspecific and overlap with those of human granulocytic anaplasmosis (caused by *Anaplasma phagocytophilum*), HFRS due to leptospirosis, or Hanta virus infection,⁴⁷¹ while lymphadenopathy has been reported as more common in SFTSV and AKI more common in Hanta virus infection (>80%).⁴⁷⁸ CD3- and CD4-positive T lymphocytes were lower than normal in patients, while natural killer cell counts were increased.⁴⁷⁹ Impaired β -interferon production and induction of a proinflammatory cytokine storm contributed to disease pathogenesis.^{464,480} Recent investigations showed that non-structural proteins of SFTSV could suppress exogenous type I interferon-induced Janus kinase/signal transducer and activator of transcription (Jak/STAT) signaling to facilitate the virus evading host immune surveillance.⁴⁸¹ Diagnosis can be made by serological testing with ELISA or detection of viral genome using reverse transcription-PCR or reverse transcription-cross-priming amplification.^{482–485} Supportive therapy and treatment of various complications are currently the mainstay of management.⁴⁶⁵

HANTAVIRUSES AND HEMORRHAGIC FEVER WITH RENAL SYNDROME

Hantaviruses are single-stranded, enveloped, negative-sense RNA viruses that belong to the Bunyaviridae family.⁴⁸⁶ Hantavirus is named after the Hantaan River in South Korea around which an outbreak occurred. The virus was isolated in the late 1970s.⁴⁸⁷ The zoonosis is endemic in East Asia, including China, Korea, and part of Russia.^{488–490} These viruses normally infect rodents, the field mouse *Apodemus agrarius* being the major animal reservoir, and do not cause disease in these hosts. Humans may become infected with hantaviruses through aerosolized excreta inhalation and contact infection by rodent urine, saliva, or feces, and human-to-human transmission has been reported with the Andes virus in South America.⁴⁹¹ Some strains of hantaviruses cause potentially fatal diseases in humans, such as hantavirus HFRS and hantavirus pulmonary syndrome – also known as hantavirus cardiopulmonary syndrome.⁴⁸⁷

HFRS (also termed Korean hemorrhagic fever, epidemic hemorrhagic fever, or nephropathia epidemica) is a group of clinically similar illnesses caused by species of hantaviruses. HFRS can thus be caused by the Hantaan virus (HTNV), Amur virus (AMV), Seoul virus (SEOV), Dobrava virus (DOBV), or Puumala virus (PUUV), with variable disease severity (often more severe with HTNV and DOBV).⁴⁹² HTNV and SEOV are found in China, with the main natural hosts being *A. agrarius* and *Rattus norvegicus*, respectively.⁴⁹³ SEOV is globally widespread and may cause moderate HFRS, whereas PUUV is endemic in northern Europe and may cause a generally mild form of HFRS, also called nephropathia epidemica. About 100,000 cases of HFRS are documented annually,⁴⁹⁴ most of which are in China, Korea, and Russia, though the disease has also been reported in Europe, Africa, and the Americas.⁴⁹⁵ PUUV is found in Scandinavia, Western Europe, and western part of Russia. DOBV is found primarily in the Balkans, and SEOV is found worldwide. Saaremaa virus is found in central Europe and Scandinavia.

Hantavirus mainly infects human vascular endothelial cells and causes extensive damage to capillaries and small vessels. Viral entry in target cells is mediated via integrins and CD55,

and pathogenic virus strains appear to utilize beta 3 integrins while beta 1 integrin is utilized by nonpathogenic strains.⁴⁹⁶

Immune-mediated mechanisms including cytokines and complement activation are involved in the pathogenesis of organ injury.^{497–499} Clinical manifestations include AKI, increased vascular permeability, and coagulation abnormalities. Typical kidney biopsy findings include acute tubulointerstitial nephritis, hemorrhage into medullary tissues, interstitial edema, renal tubular cell necrosis, and in some cases podocyte injury with foot process effacement associated with mislocalization of junctional proteins thereby resulting in glomerular proteinuria.^{500,501} While there is no gross cytopathic effect, changes of subcellular structures are accompanied by cellular dysfunction, and typical disruption of cell–cell contact, as occurs in glomerular endothelial cells.⁴⁹⁶

Data from in vitro studies also showed that hantavirus-infected vascular endothelial cells were sensitized to the action of vascular endothelial growth factor in increasing permeability.⁵⁰² AKI was associated with increased mortality, especially in elderly patients who also had more severe hypotension.⁵⁰³ In HFRS, the clinical picture is characterized by AKI, often with massive proteinuria due to tubular and glomerular involvement.⁴⁸⁶ Infection with hantavirus is associated with significant morbidity and mortality worldwide. Approximately a quarter of infected patients have a severe course. A mortality rate of 5%–10% has been reported from Korea, whereas in Europe the mortality rate appeared lower at 0.5%.⁴⁸⁶

Symptoms usually develop 1–2 weeks after exposure to infectious material, but can be delayed for weeks. Initial symptoms begin suddenly and include severe headache, back, and abdominal pain, fever, chills, nausea, facial flushing, conjunctival injection, and skin rash. Severe manifestations include hypotension, vascular leakage, and AKI.

HFRS follows a typical clinical course (Table 80.3).^{486,504} In the initial “febrile phase,” which lasts for 3–7 days, symptoms often include facial redness, fever and sweating, headache, malaise, abdominal and back pain, nonspecific respiratory symptoms, and gastrointestinal symptoms such as nausea and diarrhea. This is followed by the “hypotensive and hemorrhagic phase,” lasting for about 2 days, which is associated with increased vascular permeability. Tachycardia occurs and

Table 80.3 Typical Clinical Phases of Hemorrhagic Fever With Renal Syndrome Due to Hantavirus Infection

Phase of Illness	Time of Occurrence	Principal Features	Signs and Symptoms	Complications
1. Febrile	1–7 days	Fever	• Headache • Vomiting • Abdominal pain • Back pain • Visual disturbance	• Acute encephalomyelitis • Bleeding • Shock • Multiorgan failure • Pituitary hemorrhage
2. Hypotensive	1–3 days	Hypotension	• Capillary leak • Pulmonary symptoms	• Renal • Pulmonary edema
3. Oliguric	2–6 days	↓ urine output	• Severe oliguria can occur	• Acute respiratory distress syndrome
4. Polyuric	2 weeks	↑ urine output	• Fluid retention	
5. Convalescent	3–6 months		• Decrease of body weight • Weakness • Fatigue	• Disseminated intravascular coagulation

Modified from Jiang H, Du H, Wang LM, et al. Hemorrhagic fever with renal syndrome: Pathogenesis and clinical picture. *Front Cell Infect Microbiol.* 2016;6:1.

patients can develop hypoxemia. Leukocytosis occurs with thrombocytopenia, elevated C-reactive protein, and lactate dehydrogenase levels. The next stage is the “kidney phase,” which lasts 3–7 days, characterized by AKI with frequent oliguria, hematuria, proteinuria, and hypoalbuminemia. The severity of AKI has been associated with the severity of thrombocytopenia and also the presence of glomerular proteinuria.^{486,503,505} This initial oliguria is followed by subsequent polyuria and convalescence.

Diagnosis is by serological testing of IgM against hantavirus by various means such as reverse transcription-PCR.^{506,507} Supportive therapy is still the mainstay of care for patients with hantavirus infection.⁵⁰⁸ Careful attention to fluid management and hemodynamics is necessary, with RRT by intermittent HD or continuous RRT in unstable subjects. The role of ribavirin remains controversial, with recent results showing a lack of efficacy and safety.^{509–511}

DIALYSIS IN ASIA

Dialysis, while life-saving for patients with ESKD, is a high-cost treatment, and the access to dialysis in developing countries or those with less developed national health-care insurance systems is fraught with economic and ethical issues. It has been reported that >2.6 million people received RRT worldwide in 2010, while an estimated 2.3 million people might have died prematurely due to limited access to RRT, with the largest treatment gaps in low-income countries in Asia and Africa, which affected populations amounting to 1.9 million and 0.4 million, respectively.²⁴ Over the period of 2000–2013 the incidence of ESKD increased 13 times in Thailand, two to three times in the Philippines, Malaysia, and South Korea, and the reported figures were affected by the variable improvements in the access to RRT between different countries.⁵¹² While close to 1 million people are on maintenance dialysis in Asia, the prevalence varies considerably within the region, owing to differences in the status of development of health-care systems in different regions. Access to dialysis therapy is not restricted by personal financial means in Japan, Taiwan, South Korea, Singapore, and Hong Kong, and there is ongoing expansion of health-care insurance coverage for dialysis in other places such as Thailand, Malaysia, and many parts in China; and dialysis registries have been established in Japan, South Korea, Taiwan, Malaysia, Hong Kong, Shanghai, Thailand, and Singapore.³¹ The incidence and prevalence rates of treated ESKD are highest in Japan, Taiwan, South Korea, and Singapore, at approximately 300–450 pmp and 1800–3000 pmp, respectively, and are much lower in areas with lower income such as the Philippines and Indonesia.⁵¹² With the size of the ESKD population predicted to increase to 5.4 million by 2030, the financial implications of the RRT program are daunting.²⁴

PERITONEAL DIALYSIS

Insufficient HD facilities, draconian selection criteria for RRT, cost containment, and clinical outcomes that were at least comparable with HD in the short-term were factors that led to the establishment of a PD-First policy for publicly funded dialysis service in Hong Kong in the mid-1980s, and to this date Hong Kong continues to report the highest PD

utilization rate in the world (73% of all dialysis patients).^{513–516} A similar PD-First policy has been implemented since 2008 in Thailand, which presently has approximately 60% of dialysis patients in the public health-care system on PD and 40% on HD, and the policy has been shown to facilitate cost containment (health-care cost reduced by approximately 1500 USD per patient per year) and increase dialysis utilization by the general public (prevalence rate of ESKD increased from 420 pmp in 2007 to 1073 pmp in 2013).⁵¹⁷ A recent study from Indonesia based on economic modeling reported an equal number of 5.93 life-years saved by either a PD-First or an HD-First policy, while a PD-First policy would be associated with a slightly higher quality-adjusted life-years of 4.40 (compared with 4.34 for HD-First), a lower total lifetime treatment cost of 700 million IDR (51,800 USD) per patient (compared with 54,400 USD for HD-First), and a substantial health-care budget saving in the long term.⁵¹⁸ From a clinical and patient-focused perspective, PD would be more suitable than HD in patients with unstable cardiovascular status or vascular access problems, or those who live far from dialysis facilities. Care should be exercised to ensure adequate dialysis dose and ultrafiltration in patients of large body size, especially those with minimal residual kidney function.⁵¹⁹ The incidence rate of PD-associated peritonitis has decreased significantly consequent to improvements in connectology, with peritonitis rates of <0.3 episode per patient-year reported by some centers, below the 0.5 episode per patient-year as recommended by the International Society for Peritoneal Dialysis.^{520,521} Despite the advent of PD solutions with improved biocompatibility,^{522,523} progressive peritoneal membrane fibrosis and deterioration of dialytic functions of the peritoneum remain unresolved problems limiting the long-term application of PD and also having an adverse impact on patient outcomes.^{524–526} In addition, although the number of affected patients accounts for a relatively small percentage of all PD patients, with reported incidence rates of 0.7%–3.3%, encapsulating peritoneal sclerosis is a severe and often devastating complication with no effective therapy, which can present during dialysis or after kidney transplantation, and lead to much patient suffering often over long durations and eventually a high mortality rate of >50%.⁵²⁷ Duration of PD is a significant risk factor, because encapsulating peritoneal sclerosis predominantly affects patients on PD for >5 years, and the incidence rate increases with further time on PD.⁵²⁸ The implementation of health-care policies that place a large proportion of patients on long-term PD must therefore incorporate long-term perspectives, and measures, to minimize the occurrence of complications consequent to peritoneal fibrosis after prolonged PD, which incur not only patient suffering but also high treatment costs.

HEMODIALYSIS

The HD service in Japan is characterized by a high proportion of in-center HD (>85%), which is related to the progressive establishment of accessible HD centers over the past decades, and clinical outcomes that are deemed very favorable by international standards including both early and long-term survival rates.⁵¹² The low rate of kidney transplantation in Japan has contributed to the progressive expansion of the dialysis population, especially the elderly age group and those with diabetes mellitus.⁵²⁹ Unadjusted 5-year survival rate was

60% in Japan compared with 39% in USA in 2004–2008.⁵³⁰ Studies from the Dialysis Outcomes and Practice Patterns Study (DOPPS) were launched in the USA in 1996 and extended to Japan in 1998–2002, and to China in 2010. Results from DOPPS and Japan Dialysis Outcomes and Practice Patterns Study (J-DOPPS) showed that, compared with Western countries, HD patients in Japan showed a high utilization rate (>90%) of arteriovenous fistula, less intensive anemia management with lower target hemoglobin and lower doses of erythropoietin-stimulating agent as well as IV iron, lower parathyroid hormone levels, and lower levels of C-reactive protein; and treatment time was >4 hours in approximately 80% of Japanese patients, compared with 40% in the United States.^{512,531} Similar to earlier findings in the Hemodialysis Study and DOPPS, results from the MONDO database, which included data of HD patients from different networks globally including the Asian-Pacific region, showed that failure to convert from central venous catheter access to noncatheter vascular access within 6 months of commencing HD was associated with a higher mortality rate.^{532–534} Twice-weekly HD is not uncommon in Asian countries especially those with insufficient access to HD or where patients are required to pay for the treatment. In this regard, use of twice-weekly HD is <3% in Japan, Europe, and North America, but is approximately 20% in China.⁵³⁵ Data from serial phases of J-DOPPS also showed a U-shaped relationship between blood pressure and all-cause mortality, with lowest mortality in patients with baseline systolic blood pressure of 140–159 mm Hg and diastolic blood pressure of 65–74 mm Hg,⁵³⁶ an association between vascular access failure and high intradialytic ultrafiltration (HR: 1.41 when ultrafiltration was 5.1%–13.7% of body weight),⁵³⁷ an inverse relationship between employment and mortality or hospitalization,⁵³⁸ and improved prediction of cardiovascular events using a new model which incorporated six variables (age, diabetes mellitus, history of cardiovascular events, dialysis time per session, serum phosphorus level, and serum albumin level) compared with the Framingham Risk Score.⁵³⁹ Recent data on 1350 patients from the China Dialysis Outcomes and Practice Patterns Study (C-DOPPS) showed that 21% had hemoglobin <9 g/dL, and attributed the relatively high percentage to less frequent HD and financial barrier.⁵⁴⁰

KIDNEY TRANSPLANTATION—SPECIAL ISSUES IN ASIA

KIDNEY TRANSPLANTATION ACROSS ABO BLOOD GROUP BARRIER

Presently there are >0.3 million Japanese patients on long-term dialysis.⁴⁴ Due to the rarity of deceased-donor organ transplantation, kidney transplantation across ABO blood group barrier has developed in Japan. The first case of ABO-incompatible (ABO-I) kidney transplantation from a live donor in Japan was reported in 1991.⁵⁴¹ Subsequently >3500 ABO-I kidney transplantations have been performed, with outcomes now comparable with ABO-compatible (ABO-C) procedures.^{542–544} The success is attributed to continued improvements in preoperative desensitization therapy, which prevents posttransplant serious acute antibody-mediated rejection.^{544,545} ABO-I kidney transplantation programs are

now well established in many transplant centers in Japan, and account for approximately 30% of live-donor kidney transplant operations in Japan.⁵⁴⁴ The numbers of centers and patients undergoing ABO-I kidney transplantation have also increased rapidly in Korea, and ABO-I kidney transplantation accounted for 10% and >20% of all live-donor kidney transplantations in 2010 and 2014, respectively.^{546–548} ABO-I kidney transplantation is also performed in selected patients in Thailand, Malaysia, China, and Hong Kong.

Antiblood group antibody-mediated acute rejection ceases to appear after accommodation is induced a few weeks after kidney transplantation.⁵⁴⁹ Therefore the most important elements in the pretransplant desensitization protocol are removal of preformed antibodies and inhibition of antibody production by B cells. Previously, this could only be achieved with splenectomy.^{550,551} Rituximab has been used in ABO-I kidney transplantation in Japan since 2002, and is now an integral element in standard desensitization protocols.^{552,553} The incorporation of tacrolimus and mycophenolate into the immunosuppressive regimen has also contributed to improved clinical outcomes.⁵⁵⁴ The 5-year graft survival rates of ABO-I kidney transplantation in Japan improved from 68% in 1989–1994 to 90% in 2001–2006, and further improved after the introduction of rituximab.^{543,545} The 5-year graft survival rates of ABO-C, ABO-I with splenectomy, and ABO-I with rituximab were reported as 88.4%, 90.3%, and 100%, respectively.⁵⁵⁵ An analysis of single-center data from 1032 live-donor kidney transplantations performed in 1989–2013, of whom 247 were ABO-I, showed that ABO-I and ABO-C cases had comparable 9-year graft survival rates (86.9% and 92.0%, respectively), and that the graft survival was better in 2005–2013 compared with 1998–2004 (HR: 0.30).⁴⁸

A desensitization protocol comprising mycophenolate mofetil and low-dose corticosteroids started 28 days pretransplant, preoperative plasmapheresis, and two doses of rituximab at 375 mg/m² on day –14 and day –1, and postoperative immunosuppression with basiliximab, corticosteroids, and tacrolimus or cyclosporine was investigated in a prospective 1-year study that included 18 patients who underwent ABO-I kidney transplantation, and the results showed high tolerability of the treatment protocol and high efficacy with no anti-A/B antibody-mediated rejection.⁵⁵⁶ One patient developed anti-HLA antibody-mediated rejection, which was successfully managed with standard antirejection therapy. Patient and graft survival were both 100% after 1 year.

In a retrospective series of 327 Japanese patients who underwent live-donor kidney transplantation, comparison of protocol graft biopsies performed at 3 and 12 months after transplantation between 226 ABO-C and 101 ABO-I patients showed comparable allograft histology overall, with subclinical acute rejection detected in 6.9% and 9.9% of patients in the ABO-C and ABO-I groups at 3 months ($P = .4$) and in 12.4% and 10.1% at 12 months, respectively ($P = .5$), and no between-group difference with regard to the rate of infections (including BK nephropathy) and patient or graft survival.⁵⁵⁷

Worldwide, the Collaborative Transplant Study reported outcomes from 1420 ABO-I kidney transplantations performed at 101 transplant centers from 2005 to 2012.⁵⁵⁸ The 3-year graft survival rate of ABO-I transplants did not differ from matched ABO-C cases (89.9% vs. 90.1%). Patient survival in ABO-I cases was lower initially due to infection-related deaths,

but 3-year patient survival rates were similar between the two groups (95.6% in ABO-I vs. 96.3% in ABO-C). Data from 125 Korean patients who underwent ABO-I kidney transplantation in 2007 to 2010 showed 2-year graft and patient survival rates of 97.5% and 99.2%, respectively.⁵⁴⁶

HEPATITIS B IN KIDNEY TRANSPLANT RECIPIENTS

One-third of the world's population shows serological evidence of HBV infection, and >350 million people are chronic HBV carriers, with a worldwide seroprevalence rate of 3.61% for HBsAg.^{559–561} Endemicity of HBV infection varies between Asian countries, with HBsAg seroprevalence rates ranging from <1% to >10% (Table 80.4).⁵⁶¹

Hepatitis B vaccination and matching of HBV status between donor and recipient are measures to prevent HBV infection in kidney transplant recipients. Hepatitis B vaccination at birth was introduced in some Asian countries since the late 1980s but is still not universally practiced, especially in less developed areas. EASL clinical practice guidelines recommend vaccination in HBV seronegative patients with end-stage renal disease.⁵⁶² The efficacy of hepatitis B vaccination is significantly lower in patients with moderate to severe renal impairment and in immunosuppressed patients,^{563,564} and thus patients with CKD should receive vaccination early. Maintaining an anti-HBs titer of ≥ 100 IU/L with booster is in general recommended, although hepatitis B reactivation could still occur, albeit uncommonly, in anti-HBc-positive subjects upon heavy immunosuppression.^{565,566} Donor–recipient matching with regard to HBV serological status aims to prevent transmission of HBV from the kidney donor to the recipient. An organ from an HBsAg-positive donor must not be transplanted into a recipient who is seronegative for both HBsAg and anti-HBs except when the transplant is deemed urgent and antiviral treatment is available. The risk of HBV transmission from HBsAg-negative anti-HBc-positive donors to HBsAg-negative recipients appeared low, especially when the recipient is anti-HBs-positive,⁵⁶⁷ and transplanting a kidney from an HBsAg-positive donor into an anti-HBs-positive recipient with immunoglobulin cover appeared safe,⁵² although in both cases surveillance of de novo HBV infection with HBsAg and HBV DNA monitoring is advisable.

Immunosuppression enhances viral replication, leading to reactivation of hepatitis B and progressive liver disease.^{568,569}

Without treatment, reactivation of HBV infection occurs in patients who receive immunosuppressive agents for the treatment of diseases or the prevention of rejection after transplantation. Before the availability of antiviral prophylaxis, the rates of HBV reactivation in HBsAg-positive kidney transplant recipients were reported as 50%–94%, and HBV infection was associated with significantly increased morbidity and mortality, related to liver complications and nonliver complications, especially infections.^{50,568,570–573} Clinical manifestations of HBV-related liver complications after kidney transplantation can present as acute hepatitis, fibrosing cholestatic hepatitis, chronic hepatitis with fulminant reactivation, chronic hepatitis, cirrhosis and its complications, or hepatocellular carcinoma.^{50,51,573–578} The 10- and 20-year patient survival rates in HBsAg-positive kidney transplant recipients without antiviral therapy were 85% and 71%, respectively (vs. 98% and 95% at 10 and 20 years in HBsAg-negative patients).⁵¹ Prior to the availability of antiviral treatment, HBsAg-positive kidney transplant recipients showed a 9.7-fold increased risk of death and 68-fold increase in liver-related mortality rate compared with HBsAg-negative controls, while the survival of these patients is similar to subjects without HBV infection in the present era of effective antiviral nucleoside/tide analog therapies.⁵⁰

Interferon as a therapy for hepatitis B precipitates allograft dysfunction and rejection and is thus contraindicated in kidney allograft recipients except for life-saving indications.⁵⁷⁹ The advent of effective antiviral nucleoside/tide analog therapies has markedly improved posttransplant outcomes of patients infected with HBV. Antiviral treatment can be given prophylactically at the time of kidney transplantation or preemptively when increasing level of circulating HBV DNA indicate impending reactivation, although the former is more convenient especially in the context of a busy clinical setting. Salvage therapy, given after the onset of clinically evident liver exacerbation, is associated with inferior outcome.⁵⁸⁰ A study that included 26 HBsAg-positive Chinese patients who received lamivudine therapy preemptively based on increasing HBV DNA levels showed effective suppression of viral replication and normalization of liver biochemistry,

Table 80.4 HBsAg Seroprevalence and the Number of People Living With Chronic HBV in the General Population in Some Asian Countries

Country	Prevalence Estimates (%; 95% CI)	Population Size Per Country	HBsAg-Positive Population
China	5.49 (5.47–5.50)	1,359,821,465	74,601,204
Japan	1.02 (1.01–1.02)	127,352,833	1,294,431
Malaysia	0.74 (0.70–0.77)	28,275,835	208,540
Mongolia	9.07 (8.41–9.78)	2,712,738	246,070
The Philippines	4.63 (4.53–4.73)	93,444,322	4,326,212
Singapore	4.09 (3.87–4.33)	5,078,969	207,943
South Korea	4.36 (4.36–4.37)	48,453,931	2,111,914
Thailand	6.42 (6.37–6.47)	66,402,316	4,260,008
Vietnam	10.79 (10.29–11.31)	89,047,397	9,607,438

CI, Confidence interval; HBsAg, hepatitis B surface antigen; HBV, hepatitis B virus.

Modified from Schweitzer A, Horn J, Mikolajczyk RT, et al. Estimations of worldwide prevalence of chronic hepatitis B virus infection: a systematic review of data published between 1965 and 2013. *Lancet* 2015;386:1546–1555.

and improved survival rate to that approaching HBsAg-negative patients.⁵⁰ The high incidence of drug resistance associated with lamivudine was tackled with newer antiviral agents including adefovir, entecavir, and tenofovir.^{581,582} Adefovir is nephrotoxic and its role has been superseded by newer agents. At present, entecavir and tenofovir are the recommended first-line antiviral therapies.²⁸⁵ Entecavir is preferred in the kidney transplant setting especially in patients with suboptimal graft function as tenofovir has been associated with renal tubular toxicity.⁵⁸³ A 20-year patient survival rate of 69% has been reported for HBsAg-positive kidney transplant recipients transplanted in the recent era of antiviral treatment, which was superior to 40% in patients transplanted before the advent of antiviral drugs.⁵⁸² Overall, the short- and medium-term survival rates of HBsAg-positive kidney transplant patients are now comparable with subjects not infected with HBV.^{50,584,585} Nevertheless, while antiviral therapy prevents severe exacerbation due to heavy immunosuppression, liver-related deaths can still occur during long-term follow-up due to progressive liver disease. Data from a retrospective study of 74 Korean patients who had received antiviral treatment showed comparable renal allograft survival between HBsAg-positive and HBsAg-negative patients, while seropositivity for HBsAg was associated with 2.19-fold increased risk of death, with one-third of the deaths related to HBV, during a mean follow-up of 75.7 months.⁵⁸⁶ In another multicenter study that included 160 Korean kidney transplant recipients with HBV infection, transplanted after 1999, HBV infection was associated with 2.37-fold increased risk of death, and mortality correlated with the severity of inflammation in the prekidney transplant liver biopsy.⁵⁸⁷ Surveillance measures including monitoring of liver function and the development of malignancy by alpha-fetoprotein measurement and liver ultrasound are thus important elements in the clinical care of kidney transplant recipients infected with HBV in Asia.

B-cell depletion with rituximab is often included in desensitizing immunosuppressive protocols for patients with antituberculous group or anti-HLA antibodies prior to kidney transplantation.⁵⁸⁸ However, rituximab is associated with a high rate of hepatitis B reactivation, not only in HBsAg-positive subjects but also in subjects with apparent clinical resolution of prior infection as denoted conventionally by seronegativity for HBsAg and seropositivity for anti-HBc.^{589–592} The rate of hepatitis B reactivation in HBsAg-negative/anti-HBc-positive patients who received rituximab-containing chemotherapy was reported as 11.3%–41.5%.^{593–595} Patients with autoimmune diseases treated with rituximab are similarly at risk.^{596,597} In this regard, a study in 49 HBsAg-negative anti-HBc-positive kidney transplant recipients in Korea showed hepatitis B reactivation in five patients (10.2%), in whom two were regarded clinically severe, resulting in one mortality.⁵⁹⁸

Prior to the advent of effective and safe antiviral treatment, patients with cirrhosis were largely declined for kidney transplantation in view of the unacceptably high rate of liver-related mortality.⁵⁹⁹ In the present era patients with compensated cirrhosis and no evidence of portal hypertension can be considered for kidney transplantation. Data from a retrospective study in Korea of 12 kidney transplant recipients with prior HBV-associated liver cirrhosis showed liver deterioration after kidney transplantation in one patient and stable liver status in the other 11 patients.⁶⁰⁰ Patient survival rate at 5 years was similar in patients with or without

prior cirrhosis, at 100% and 94%, respectively. Patients with decompensated cirrhosis and those with compensated cirrhosis and portal hypertension are not eligible for kidney transplantation, but may be considered for combined liver and kidney transplantation.⁵⁹⁹

CONCLUDING STATEMENTS

Clinical nephrology has advanced significantly over the past half a century in the Far East. Variations in geotopography and genetic background, and rapid changes in the urban environment and socioeconomic development, have contributed to the marked diversity of disease spectra and multiplicity of challenges to the prevention and treatment of kidney diseases in the region. By contrast, the relatively low facility costs in many parts of Asia and the abundance of patients and biomedical samples present invaluable research opportunities to address knowledge gaps that are critical in the strive to continue to improve clinical outcomes.



Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. Mr Chan is a 29-year-old Chinese male with good past health. He presents with malaise and ankle swelling for 2 weeks. Physical examination shows pallor and bilateral pitting ankle edema. Blood pressure reading is 130/90 mm Hg. Urine is 3+ for protein and microscopy shows numerous red blood cells and scanty white blood cells. Serum creatinine level is 1.02 mg/dL, which is within the normal range. Serum albumin is below normal, at 29 g/L. Other investigation results include hemoglobin 8.3 g/dL, white cell count $3.2 \times 10^9/L$, and platelet count $86 \times 10^9/L$.

Which one of the following tests is most relevant?

- Blood test for antidouble-stranded DNA (dsDNA) antibodies
- Blood test for antiphospholipase A2 receptor antibody
- Blood test for antineutrophil cytoplasm antibodies
- Blood test for HbA1c level
- Blood test for cholesterol level

Answer: a

Rationale: Lupus nephritis is the leading cause of secondary glomerular disease presenting as acute glomerulonephritis in Asian populations. Although more common in females at a female-to-male ratio of 8–9 to 1, it is not uncommon to see male Asian patients with lupus nephritis. Indeed, renal involvement is more common in male lupus patients compared with females. In addition to the renal abnormalities, the presence of thrombocytopenia, low white cell count, and anemia points toward lupus being the underlying diagnosis.

2. Mrs. Lee is a 63-year-old Chinese female who has emigrated to the United States from Mainland China a year ago. She has a history of hypertension for 3 years, for which she has been taking losartan 50 mg daily. There is no previous medical investigation. She presents with malaise over the past 4 months. Physical examination shows mild pallor and no ankle edema. Blood pressure reading is 130/90 mm Hg. Urine shows trace proteinuria and microscopy shows scanty red blood cells and white blood cells. There are also dysmorphic cells in urine cytology examination. Her serum creatinine level is 2.83 mg/dL and estimated glomerular filtration rate is 17 mL/min/1.73 m² by the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation. Further questioning reveals that she has a habit of taking herbal medicine on a regular basis. Other results include hemoglobin 8.3 g/dL, white cell count $6.2 \times 10^9/L$, and platelet count $186 \times 10^9/L$. Ultrasound examination shows reduced kidney size, with both sides measuring approximately 7 cm in length.

Which one of the following is the proper course of action?

- She probably has hypertensive nephrosclerosis. Therefore one should titrate her medications to aim for a lower blood pressure; arrange serial monitoring of her kidney function, and explain to her that she has chronic kidney disease and progressive deterioration of kidney function is the likely clinical course.
- Continue with the present dose of losartan; arrange serial monitoring of her kidney function, and explain to her that she has chronic kidney disease and progressive deterioration of kidney function is the likely clinical course.

- Titrate her medications to aim for a lower blood pressure; arrange kidney biopsy; arrange serial monitoring of her kidney function, and explain to her that she has chronic kidney disease and progressive deterioration of kidney function is the likely clinical course.
- Titrate her medications to aim for a lower blood pressure; arrange serial monitoring of her kidney function, and explain to her that she has chronic kidney disease and progressive deterioration of kidney function is the likely clinical course; arrange investigations to look for urothelial malignancies.
- Continue with the present dose of losartan; arrange serial monitoring of her kidney function, and explain to her that she has chronic kidney disease and progressive deterioration of kidney function is the likely clinical course; arrange investigations to look for urothelial malignancies.

Answer: d

Rationale: Investigation results show that this woman has chronic kidney disease, and her current blood pressure is too high. In addition, the history indicates that she has regular exposure to herbal medications, and the urine cytology result raises the suspicion of urothelial malignancy. The overall picture can be compatible with aristolochic acid nephropathy, in which multifocal urothelial malignancy can occur in addition to chronic kidney disease.

3. A 43-year-old male presents with high swinging fever and weakness for 2 days. A week ago he returned from a trip to Taiwan, which lasted 2 weeks during which he had visited rice farms and had stayed in farmhouses. Flulike symptoms began 3–4 days ago, followed by increasing prostration and fever. There is also headache and myalgia, and he has noticed decreased urine output since yesterday. Physical examination shows fever at 39°C, no palpable lymphadenopathy, no respiratory abnormality, and both the liver and spleen are not palpable. Urine shows trace proteinuria and microscopy shows no abnormality. Other investigation results include serum creatinine level 2.83 mg/dL [estimated glomerular filtration rate is 26.1 mL/min/1.73 m² by the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation], hemoglobin 14.2 g/dL, white cell count $14.2 \times 10^9/L$, platelet count $116 \times 10^9/L$, alanine aminotransferase 600 IU/L, bilirubin 32 mg/dL, alkaline phosphatase 180 U/L, and creatine kinase 190 U/L which is just above the upper limit of normal. Chest X-ray is clear. Ultrasound examination shows normal kidney size and no abnormality; and there was no feature to suggest obstruction in the biliary tract. Which one of the following is the proper course of action?

- Arrange thin and thick blood smear to look for malaria immediately.
- Institute forced alkaline diuresis for the treatment of rhabdomyolysis resulting in acute kidney injury.
- Arrange IgM antibody test for dengue fever.
- Arrange antibody or polymerase chain reaction (PCR) test for leptospirosis and start treatment with intravenous penicillin.
- Arrange antibody or PCR test for scrub typhus due to *Orientia tsutsugamushi*.

Answer: d

Rationale: Features of hemorrhagic fever, which could occur in hantavirus infection or dengue hemorrhagic fever, are absent in this man. Skin rash, with or without eschar, is usually present in scrub typhus. In addition to acute kidney injury, the high bilirubin level and thrombocytopenia are suggestive of

leptospirosis. *Leptospira* enters the human body through the cutaneous or mucosal route or by aerosolization. Rodents and small mammals are reservoirs, and infected animals shed the spirochete in their urine into water or soil. Exposure to sewage or farm areas with rodent infestation and an incubation period of 5–14 days are relevant parts in the history.